

OMNIVCHAT: SYNTHESIZING, BENCHMARKING, AND TRAINING FOR NATIVE AUDIO-VISUAL DIALOGUE

Haolin He^{1,2,*}, Yunfei Chu^{2,*}, Qi Chen^{2,3,4,*}, Wen Huang^{2,3}, Yuan Feng², Muzhi Zhu^{2,5},
Zheqi Dai¹, Haoning Xu¹, Dongchao Yang^{1,2}, Chunyat Wu¹, Zining Liang¹, Zhengxi Liu¹,
Xiquan Li³, Xie Chen³, Xize Cheng², Qize Yang², Jin Xu^{2,†}, Qiuqiang Kong^{1,†}

¹The Chinese University of Hong Kong ²Alibaba Token Hub, Alibaba Group
³Shanghai Jiao Tong University ⁴Shanghai Innovation Institute ⁵Zhejiang University

ABSTRACT

We define **OmniVChat** (Omni Video Chat) as the task of **native audio-visual dialogue** between a user and an omni model. In OmniVChat, omni models directly and simultaneously receive audio and video from a user and return text. The user’s query is embedded in the audio and video, without a separate text question, external captioning, or speech recognition. Direct audio-visual input reduces external latency and computation while preserving perceptual cues. However, research on OmniVChat faces two constraints: data availability and evaluation. Recordings of people using their own devices are scarce. Furthermore, a good reply often needs to account for the user’s surroundings, facial expressions, and nearby objects, and such responses can be expressed in many different ways, making keyword matching unreliable for evaluating reply quality. Recent progress in agent systems and video generation makes **generation for comprehension** viable, which means using synthesized dialogues for training and evaluation. Therefore, we present **OmniVChat-Studio**, a multi-agent data engine for synthesizing single- and multi-turn audio-visual dialogues. We use synthesized dialogues to build **OmniVChat-Bench**, an evaluation benchmark that evaluates omni models’ basic dialogue abilities across five ability categories. We also present **OmniVChat-RL**, a reinforcement learning reward design that jointly targets reply correctness, efficiency, and style in OmniVChat. Training Qwen3-Omni-Instruct with OmniVChat-RL on synthesized dialogues improves its performance on both OmniVChat-Bench and the human-recorded **OmniVChat-Bench-Human**. These gains validate the reward design and show transfer to real-world dialogues in training and evaluation.

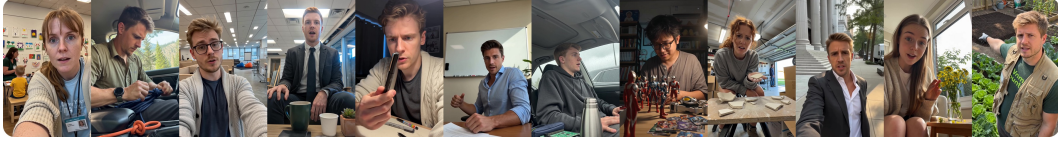


Figure 1: A glance at **OmniVChat-Bench**: each card shows one user frame from a synthesized dialogue. The benchmark spans 5 ability categories, 17 subcategories, and 22 scenario domains.

1 INTRODUCTION

We define **OmniVChat** (Omni Video Chat) as the task of **native audio-visual dialogue** between a user and an omni model (Jiang et al., 2025). In OmniVChat, omni models directly and simultaneously receive audio and video from a user and return text. The user’s query is embedded in both inputs, without a separate text question or external captioning or automatic speech recognition (ASR) (Prabhavalkar et al., 2023). Direct audio-visual input reduces external latency and computation while preserving acoustic and visual cues. However, research on OmniVChat faces two con-

*Equal contribution. [†]Corresponding authors. Work done as an intern at Qwen Team, Alibaba Token Hub.

straints: data availability and evaluation. The first challenge is limited data availability. Open-source recordings of people using their own devices are scarce. Noise, camera motion, and device posture create a long tail of input conditions. The second challenge is evaluating reply quality. Dialogues contain complex scenes and many rare cases, where a good reply should account for rich multi-modal context such as the user’s environment, facial expressions, emotion, nearby objects and so on. Because such information can be expressed in various ways, it is difficult to reliably assess reply quality through keyword matching or other rule-based methods. Few datasets and benchmarks target this native dialogue (Pourreza et al., 2026; Lu et al., 2026). Models thus tend to rely on external components such as ASR cascades (Yao et al., 2026). While practical, these designs inevitably omit subtle prosody and vocal affect, and add latency, computation, and potential transcription errors.

Progress in agent systems (Dorri et al., 2018; Yang et al., 2025) and video generation (Xing et al., 2024; Ma et al., 2025) makes **generation for comprehension** possible. This means using synthesized dialogues for training and evaluation. Our contributions are as follows. First, we present **OmniVChat-Studio**, a controllable and extensible multi-agent data engine for synthesizing native audio-visual dialogue data at scale for training and evaluation. It contains four agents that synthesize single- and multi-turn dialogues using a configuration for each subcategory. Second, we introduce **OmniVChat-Bench**, an evaluation benchmark that evaluates omni models’ basic dialogue abilities across five ability categories. Each instance includes a rubric. A rubric contains several tiers, and each tier has one or more criteria. A large language model checks whether the reply meets each criterion. Tier 0 checks whether the reply uses the user’s language. It gives no points, and failing it makes the rubric score zero. In later tiers, a criterion earns one point only after every criterion in all earlier tiers is met. Example 1 below shows how these rules apply to one instance. Third, we present **OmniVChat-RL**, a reinforcement learning (RL) reward design that jointly targets reply correctness, efficiency, and style in OmniVChat (Kaelbling et al., 1996). A rubric reward for reply correctness is first introduced using the same scoring method as in evaluation, followed by two additional reward terms for efficiency and style. **OmniVChat-Bench-Train** serves as the training corpus synthesized by OmniVChat-Studio for OmniVChat-RL, whereas **OmniVChat-Bench-Human** provides a manually recorded benchmark for testing transfer to real-world dialogues. Fine-tuning Qwen3-Omni-Instruct (Xu et al., 2025b) with OmniVChat-RL on OmniVChat-Bench-Train yields consistent performance gains across both OmniVChat-Bench and OmniVChat-Bench-Human, while simultaneously making responses more concise. These gains validate the OmniVChat-RL and demonstrate successful transfer to real-world dialogues. Given that the two benchmarks overlap across single-turn subcategories, consistent gains in these shared areas suggest that the synthetic and recorded dialogue distributions are well aligned.

The paper is organized as follows. Section 2 describes the OmniVChat-Studio framework and its single-turn and multi-turn subsystems. Section 3 presents OmniVChat-Bench for evaluating basic dialogue abilities. Section 4 presents OmniVChat-RL and the training results. Appendix B provides an overview of the related work.

2 OMNIVCHAT-STUDIO: SYNTHESIZING DIALOGUES FROM CORPUS

We present **OmniVChat-Studio**, a multi-agent data engine that takes a text corpus as input and synthesizes native audio-visual dialogue data. The resulting data contain rendered audio-visual clips, the corresponding reference replies, and rubrics for scoring model replies. OmniVChat-Studio consists of two subsystems, as shown in Figure 2. The single-turn subsystem on the left synthesizes one-turn dialogue data. The multi-turn subsystem on the right synthesizes a sequence of dependent turns, where later turns use accepted earlier clips and dialogue as history.

Both subsystems use the four agents shown in the lower-left legend of Figure 2. Each module has the same color as the agent responsible for it. The **Director** handles all text-based input and output tasks. The **Renderer** turns an accepted prompt and any media references into a synchronized audio-visual clip. The **Reviewer** examines the rendered audio and video, writes a caption and quality report, and answers focused questions when more evidence is needed. The deterministic **Validator** checks structured scripts or dialogue records against the configured rules and returns either a pass or a list of rule violations.

When synthesizing data for a new subcategory, such as multi-speaker dialogue, only the **Flexible** modules need to be designed and customized with subcategory-specific prompts (shown as rounded

Example 1: Tiered Rubric for a Festive Mug Dialogue.

Audio-Visual Input	Audio: The user says, “I love how festive everything looks tonight. Hey, the mug on the coffee table — what pattern does it have on it?” Video: A white mug on the coffee table has a red reindeer silhouette.
Reference Reply	“That mug has a red reindeer silhouette pattern on it — very festive! It goes perfectly with the rest of your Christmas setup.”
Tier 0	Criterion 1: The reply is in English.
Tier 1	Criterion 1: The reply identifies the pattern on the mug as a reindeer.
Tier 2	Criterion 1: The reply says that the pattern is red. Criterion 2: The reply says that the reindeer is a silhouette.
Tier 3	Criterion 1: The reply uses a natural and relaxed tone that fits the festive setting.
Scoring	Tier 0 adds no points, but failing it gives a final score of zero. If Tiers 0 and 1 are met, each met criterion in Tier 2 earns one point. A reply can earn one Tier 2 point even if it misses the other Tier 2 criterion. Tier 3 can add one point only after both Tier 2 criteria are met.

boxes in Figure 2). In contrast, the **Fixed** modules (shown as boxes with square corners) remain completely unchanged across all categories. The overall framework and the role of each module in both subsystems also remain consistent across subcategories.

Each module has the same circled identifier beside its name in this section, Figure 2, and the corresponding rows of Examples 2 and 3. Circle 0 marks the sampling module shared by both subsystems. Circles 1–14 label the single-turn modules on the left of Figure 2, and circles a–n label the multi-turn modules on the right. Appendix C gives the design rationale, sampling scheme, and module-level checks and associated repair procedures.

2.1 SINGLE-TURN SUBSYSTEM

The single-turn subsystem synthesizes one-turn dialogue data. Example 2 shows the corresponding outputs and decisions from one real run.

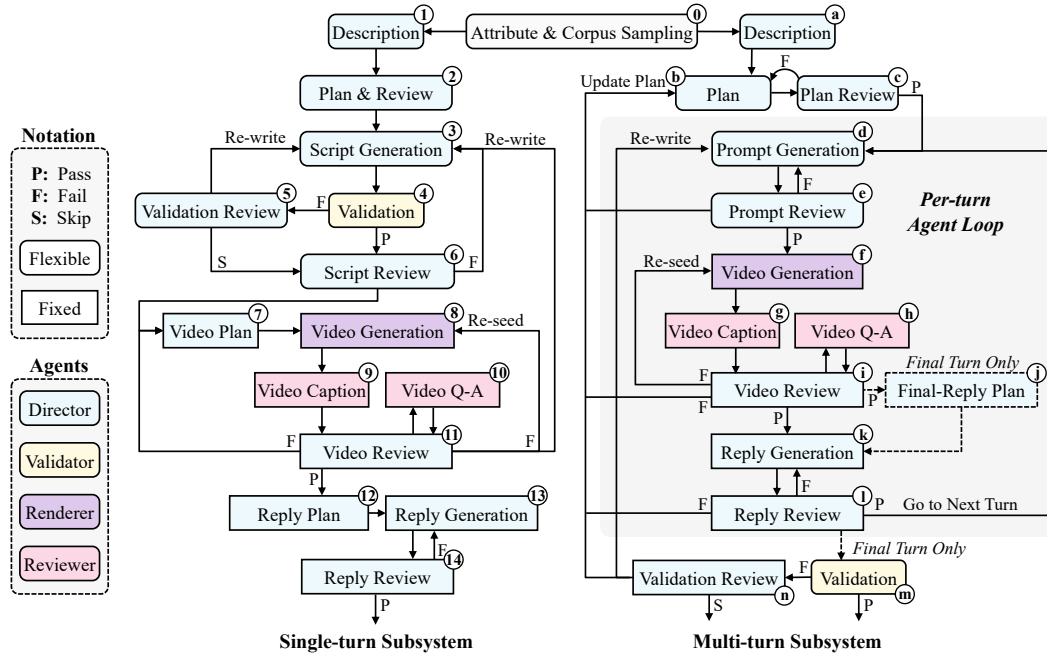


Figure 2: **OmniVChat-Studio** framework with single-turn (left) and multi-turn (right) subsystems. Each module uses the same color as the agent responsible for it.

Example 2: Summarized Module-by-Module Single-Turn Trace (Instance 085b292a).

The subcategory is *Single-turn Visual Mode Recognition (DSLIP-VDMC-SVMR)*, which tests user-relative left and right under the current camera view.

Attribute & Corpus Sampling ①	Configuration (base probabilities): Chinese/English, 40%/60%; one user, 100%; 22 scenario domains, 1/22 each; 14 camera setups, 1/14 each; audio and visual interference (none/low/medium/high), 25% each. User-question types: side identification (asking which side a target is on) and two-sided comparison (asking about differences between objects on the two sides), 1/2 each. Attribute sampling selects an English-speaking solo user, an office, a walking selfie, medium audio noise, high visual interference, and a two-sided comparison. Corpus sampling supplies three passages, including one on health.
Description ①	The Director uses the health passage to outline a corridor dialogue with two meeting-room doors and a question about the user’s left door.
Plan & Review ②	The Director refines the corridor scene and reviews the left-right mapping, recommending clear door views and shorter speech.
Script Generation ③	The Director turns the plan into an 11.3-second script with timed sections for walking, speech, and a silent wait. The user asks, “There’s a meeting room on each side of me — is the one on my left open or closed?”
Validation ④	The script passes the configured checks, including format and timing.
Validation Review ⑤	Validation ④ reports no violation, so this module is skipped.
Script Review ⑥	The Director accepts the script as a test of user-relative left and right.
Video Plan ⑦	The Director selects shared rendering instructions for the camera, a continuous shot, realism, and the absence of subtitles.
Video Generation ⑧	The Renderer produces the 11.1-second audio-visual clip from the assembled prompt.
Video Caption ⑨	The Reviewer describes the user, corridor, and speech, and checks seven aspects of video quality. The report finds no clear defect.
Video Review ⑪	The caption leaves the door states unclear, so the Director requests focused Video Q-A ⑩.
Video Q-A ⑩	The Reviewer confirms that the door on image right is closed and the door on image left is open. The Director then accepts the clip in Video Review ⑪.
Reply Plan ⑫	The Director plans a brief English reply using the door states confirmed by Video Q-A ⑩.
Reply Generation ⑬	“The one on your left is closed. The door on your right is open.” The Director also creates a tiered rubric covering language, the left door’s state, brevity, and the right door’s state.
Reply Review ⑭	The Director checks the reply and rubric against the video evidence and accepts them. The clip, reply, and rubric form the completed instance.

Attribute & Corpus Sampling ① samples attribute values and corpus passages for the current instance based on the subcategory configuration and text corpus. The configuration defines the desired probability distribution for each attribute. The corpus contains a large collection of passages, and the module samples three by default. In Example 2, circle 0 selects English, an office, and a walking selfie, along with three passages including one on health. Appendix C.2.1 gives the sampling formula, its full notation, and details on coverage and cost. **Description** ① develops a rough storyline based on the sampled corpus passages and preset attributes. In Example 2, circle 1 uses the health passage to outline a corridor dialogue about the user’s left door.

Plan & Review ② refines the key storyline and audio-visual scene details based on the scene description and sampled attributes. In Example 2, circle 2 plans how to show both doors clearly and checks the left–right mapping. **Script Generation** ③ writes a structured script based on the reviewed plan. The script contains scene fields and a timed sequence of speech, sounds, music, and on-screen text. In Example 2, circle 3 turns the corridor plan into a timed script.

Validation ④ checks for rule violations based on the synthesized script and subcategory configuration. It checks requirements such as script format and timing, then reports a pass or a list of violations. **Validation Review** ⑤ decides whether script corrections are needed based on the reported violations and the script context. This conditional module either dismisses false reports or requests script corrections. In Example 2, circle 4 reports no violations, so circle 5 is skipped. **Script Review** ⑥ evaluates the planned dialogue based on the script after the preceding checks. It checks

whether the script meets the subcategory requirements, then accepts it, requests a rewrite, or discards the run. In Example 2, circle 6 accepts the script as a test of user-relative left and right.

Video Plan^⑦ prepares the rendering prompt based on the accepted script. The Director combines the planned scene and events with compatible instructions for the camera, realism, and other shared requirements. In Example 2, circle 7 selects instructions for a continuous shot, realism, and no subtitles. **Video Generation**^⑧ renders a synchronized audio-visual clip based on the accepted rendering prompt. In Example 2, circle 8 renders the office clip.

Video Caption^⑨ writes a caption and a seven-part quality report based on the synthesized audio-visual clip. The report covers cuts, realism, delivery speed, body structure, appearance, artifacts, and on-screen text. **Video Review**^⑩ compares the rendered content with the accepted script based on the caption, quality report, and any Q-A evidence. It accepts the clip, requests a re-seed, rewrite, or focused Q-A check, or discards the run. **Video Q-A**^⑪ answers focused questions from **Video Review**^⑩ based on the rendered audio and video. This conditional module runs when requested. The caption and Q-A evidence may reveal that some scripted background details are absent or different in the clip. **Video Review**^⑩ can still accept the clip if these differences do not affect the main storyline or intended dialogue. In Example 2, circle 9 finds no clear defect, but circle 11 requests a closer check of the doors. Circle 10 confirms their states before circle 11 accepts the clip.

Reply Plan^⑫ plans a grounded reply based on the accepted audio-visual evidence. It leaves out scripted details that are not supported by the caption or Q-A evidence. **Reply Generation**^⑬ writes a reference reply and tiered rubric in the user’s language based on the reply plan and accepted audio-visual evidence. **Reply Review**^⑭ checks the synthesized reply and rubric based on the accepted audio-visual evidence and subcategory requirements. It accepts them or requests one refinement after checking language, grounding, tier order, and spoken style. Separate budgets limit each repair route, and the subsystem discards any run without a valid script, clip, reply, or rubric. In Example 2, circle 12 plans a short answer from the confirmed door states, circle 13 writes the reference reply and rubric, and circle 14 accepts both. The completed instance contains the reference reply, rubric, and a 1080P clip with 44.1 kHz stereo audio.

2.2 MULTI-TURN SUBSYSTEM

The multi-turn subsystem handles dependencies across turns. This subsection focuses on its main differences from the single-turn subsystem. The right side of Figure 2 shows these changes, and Example 3 in Appendix C.2.2 matches the modules circle by circle in one real run.

Unlike the single-turn **Plan & Review**^②, the multi-turn subsystem separates **Plan**^⑥ from **Plan Review**^⑤. **Plan**^⑥ expands the storyline into a joint plan that specifies every turn and selects the earlier clips, audio, or final frames used by later turns within a reference budget. **Plan Review**^⑤ checks all turns and media links in a separate Director call, then accepts the plan, requests revision, or discards the run. In Example 3, circles b and c plan the return to the same room, then correct the language, media link, and final silent wait.

The multi-turn subsystem also replaces the single-turn **Script Generation**^③ and **Video Plan**^⑦ with per-turn **Prompt Generation**^④. This module writes a rendering prompt for each turn based on the reviewed joint plan and shared rendering instructions. It uses the available media identifiers as placeholders for earlier clips, audio, or final frames. **Prompt Review**^⑥ checks the prompt and its media links before rendering, then accepts it, requests a local rewrite, or requests a plan update. In Example 3, circle d links the Turn 2 prompt to the Turn 1 clip, while circle e repairs the denser Turn 1 prompt and accepts the Turn 2 prompt.

3 OMNIVCHAT-BENCH: EVALUATING BASIC DIALOGUE ABILITIES

We introduce **OmniVChat-Bench**, an evaluation benchmark that evaluates omni models’ basic dialogue abilities across five ability categories. Its 2,550 single-turn and 250 multi-turn instances cover 17 subcategories and 22 scenario domains, such as Parenting, Travel and Sightseeing, and Medicine and Health. Since few public recordings cover these dialogues, OmniVChat-Studio synthesizes all

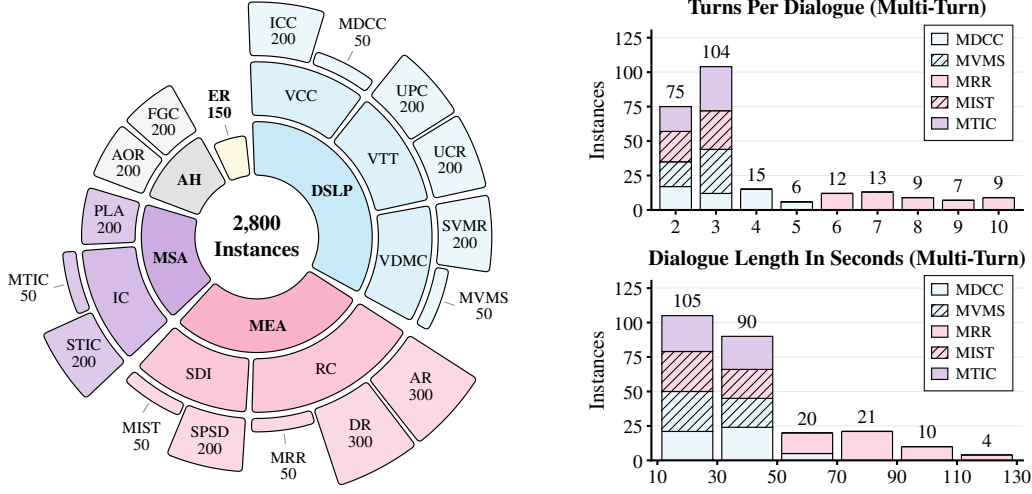


Figure 3: **Overview of OmniVChat-Bench**, with 2,800 synthesized instances. **Left:** The five ability categories and their 17 subcategories. Bar length shows the instance count from the base ring. Angular span shows the number of subcategories in each band. **Right:** Turn and duration distributions for 250 multi-turn instances, stacked by subcategory.

2,800 instances with reference replies and tiered rubrics. They include 1,766 English dialogues (63.1%) and 1,034 Chinese dialogues (36.9%). A scenario domain describes what a dialogue is about, while a subcategory specifies the basic dialogue ability under evaluation. Audio checks identify clicks and abrupt endings for repair. Human inspectors follow the inspection procedure in Appendix D.1.1 to check that each clip shows the intended dialogue. Figure 3 shows the instance distribution across ability categories and subcategories. Appendix D.5 defines the subcategories, and Appendix F.1 gives synthetic examples with reference replies and rubrics.

As in Example 1, each instance includes a reference reply and a tiered rubric. For evaluation, `qwen3.7-max` (Qwen Team, 2026a) judges whether each criterion is met. Appendix D.3 compares grader agreement and score variation. A gate controls whether a later tier can earn credit. Tier 0 checks whether the reply uses the user’s language; it earns no points, and failure makes the rubric score zero. A met criterion outside Tier 0 earns one point only when all criteria in earlier tiers are met. An incomplete tier retains its earned points but blocks later tiers. The rubric score is the number of credited points divided by the total number of criteria outside Tier 0. For each instance, let $\mathcal{K}_t$ be the set of criteria in tier $t = 0, \dots, T$, where T is the final tier. Let $\mathcal{H}(y)$ be the criteria that the grader finds met by reply y . The single-instance scoring function is:

$$r(y) = \frac{\sum_{t=1}^T |\mathcal{K}_t \cap \mathcal{H}(y)| \prod_{s=0}^{t-1} 1[\mathcal{K}_s \subseteq \mathcal{H}(y)]}{\sum_{t=1}^T |\mathcal{K}_t|}. \quad (1)$$

Here, $|B|$ counts the elements of a set B , and $1[\cdot]$ is 1 when its condition is true and 0 otherwise. The intersection counts met criteria, while the product permits credit only when all earlier tiers are complete. The denominator counts every criterion outside Tier 0, including those blocked by an incomplete earlier rubric tier, so $r(y) \in [0, 1]$.

For multi-turn evaluation, all models receive the same earlier clips and reference replies; only the final reply is scored. For an evaluation set $\mathcal{E}$, let y_i be the reply scored for instance i and $\mathcal{E}_c$ contain the instances in subcategory c . Pooled Mean gives each instance equal weight, while Subcategory Mean first averages within each subcategory and then gives all C subcategories equal weight. OmniVChat-Bench reports Subcategory Mean as its overall Mean score, with $C = 17$:

$$\text{PooledMean}(\mathcal{E}) = \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} r(y_i), \quad \text{SubcategoryMean}(\mathcal{E}) = \frac{1}{C} \sum_{c=1}^C \text{PooledMean}(\mathcal{E}_c). \quad (2)$$

Section 4.3 and Table 1 report transfer and model comparisons.

Dialogue-State & Link Perception (DSL) evaluates whether a model understands the current state of a dialogue. It covers connection status, turn boundaries, and camera orientation. For example, the model should wait when a user pauses before finishing a request and interpret left and right from the current camera view. **Multimodal Entity Alignment** (MEA) evaluates whether a model connects speech with the correct visible objects and speakers. For example, the model should identify the cup indicated by a gesture and address the active user when several people are speaking. **Model Self-Awareness** (MSA) evaluates whether a model states its identity and physical limitations correctly. For example, it should not claim that it can move an object, but it can explain what the user can do instead. **Anti-Hallucination** (AH) evaluates whether replies remain grounded in the available audio and visual evidence. For example, the model should say when a requested object is absent or correct a false description of a visible object’s color. **Emotion Recognition** (ER) evaluates whether a reply uses facial and vocal emotion cues. For example, a worried expression and tone should lead to a reply that addresses both the user’s emotion and request. Appendix D.4 compares model scores on matched single-turn and multi-turn subcategories. The recorded probe contains only single-turn dialogues and does not test live interruption; Appendix D.2 describes the procedures for both recording and annotating these dialogues.

4 OMNIChat-RL: REWARDS FOR BETTER DIALOGUE

We present **OmniVChat-RL**, an RL reward design that jointly targets reply correctness, efficiency, and style in OmniVChat. These three reply-quality signals also define the evaluation metrics. An auxiliary format term is used only during training to require a valid reply. Training and evaluation use separate dialogues from the same subcategory configurations. OmniVChat-Bench measures training gains, while recorded dialogues test transfer. Matched gains can indirectly suggest aligned distributions across the shared single-turn subcategories (§4.3).

4.1 TRAINING SETUP

To support training and test generalization to real-world dialogues, we introduce two datasets. **OmniVChat-Bench-Train** contains 5,600 training dialogues and a separate 560-dialogue development split. Each dialogue and its rubric are synthesized by OmniVChat-Studio. Development scores select the checkpoint. **OmniVChat-Bench-Human** contains 360 human-recorded single-turn dialogues for testing transfer to real-world dialogues (recording and annotation procedures in Appendix D.2). It is used only for evaluation. The base model is the Thinker of Qwen3-Omni-30B-A3B-Instruct. It has video and audio encoders and a text decoder.

Training uses group sequence policy optimization (GSPO) (Zheng et al., 2025) with the Thinker as the response policy. For each training input x , the sampling policy $\pi_{\theta_{\text{old}}}$ draws N replies $\mathcal{G}(x) = \{y_1, \dots, y_N\}$. Following the group baseline in group relative policy optimization (GRPO) (Shao et al., 2024), the advantage $A_i = R(y_i) - \mu_{\mathcal{G}}$ subtracts the group’s mean reward $\mu_{\mathcal{G}}$ from each reply’s total reward $R(y_i)$, without standard-deviation normalization. The current policy π_{θ} maximizes:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \min(q_i(\theta) A_i, \text{clip}(q_i(\theta), 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}}) A_i) \right]. \quad (3)$$

The expectation averages over training inputs and sampled replies. The full-reply probability ratio $q_i(\theta) = [\pi_{\theta}(y_i | x) / \pi_{\theta_{\text{old}}}(y_i | x)]^{1/|y_i|}$ is normalized by the reply’s token count $|y_i|$ to reduce length dependence. Clipping bounds the ratio to $[1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}}]$, and the minimum limits the incentive for large policy changes. Training uses low-rank adaptation (LoRA) (Hu et al., 2021) with rank-64 adapters for 1,000 iterations. Each iteration uses 32 inputs with four sampled replies per input, and every twentieth iteration saves a checkpoint. Appendix E.1 gives the full training settings.

4.2 REWARDS AND EVALUATION METRICS

Three quality signals connect training with evaluation: rubric correctness, reply efficiency, and style. OmniVChat-RL uses their reward forms during training, while evaluation reports the corresponding

metrics. The total training reward for reply y is:

$$R(y) = r(y) + \lambda_{\text{fmt}} f(y) + \lambda_{\text{eff}} e(y) + \lambda_{\text{sty}} s(y) \quad (4)$$

The functions $r(y)$, $f(y)$, $e(y)$, and $s(y)$ score correctness, format, efficiency, and style. The correctness reward directly uses the single-instance score $r(y)$ in Equation 1. Each training dialogue supplies a rubric with criteria for an open-ended reply (Gunjal et al., 2026). The training-only format term $f(y)$ equals one for a nonempty reply with no thinking tags and zero otherwise.

The reply efficiency reward and metric measure rubric credit relative to reply length. For training, the raw per-reply efficiency score $\rho(y)$ divides rubric credit by counted reply length. Among replies with the same positive rubric score and positive counted length, fewer words give higher raw efficiency:

$$\rho(y) = \frac{r(y)}{\max\{w(y), 1\}}, \quad (5)$$

The function $w(y)$ counts the length of reply y , not the model’s tokens. Each Chinese character counts as one unit, as does each continuous group of English letters or digits. Apostrophes and hyphens within a group do not split it, and other punctuation adds no units. The denominator $\max\{w(y), 1\}$ prevents division by zero. A reply with zero rubric credit has zero raw efficiency regardless of its length. The efficiency reward $e(y_i)$ compares the N replies sampled for the same input x . The minimum and maximum below range over those replies in $\mathcal{G}(x)$:

$$e(y_i) = \begin{cases} \frac{\rho(y_i) - \min_j \rho(y_j)}{\max_j \rho(y_j) - \min_j \rho(y_j)}, & \max_j \rho(y_j) > \min_j \rho(y_j), \\ \frac{1}{2}, & \text{otherwise,} \end{cases} \quad j \in \{1, \dots, N\}. \quad (6)$$

When the efficiencies differ, the lowest value receives reward 0 and the highest receives reward 1. The other values are placed proportionally between 0 and 1. When all values are equal, every reply receives $\frac{1}{2}$. This shared contribution cancels when the group-mean reward is subtracted to compute the advantage A_i (§4.1). The comparison stays within one input, so no fixed reply-length target is required. Evaluation reports Reply Efficiency on a set $\mathcal{E}$ as $\text{RE}(\mathcal{E}) = 1000 \sum_{y \in \mathcal{E}} r(y) / \sum_{y \in \mathcal{E}} w(y)$. This metric is the rubric credit earned per thousand counted words.

The training system prompt specifies natural speech, concise wording, and a clear distinction between roleplay and real facts. These rules and a grammar requirement form seven criteria in the style-grading prompt. The style reward $s(y)$ is one if `qwen3.7-max` confirms all seven criteria and zero otherwise. Evaluation uses the same grader and criteria, reporting Style as the mean of $s(y)$ over successfully scored replies. Appendix E.4 gives the policy and style-grading prompts; Appendix E.1 gives grader settings. The format, efficiency, and style weights are $\lambda_{\text{fmt}} = 0.5$, $\lambda_{\text{eff}} = 0.1$, and $\lambda_{\text{sty}} = 0.5$. Correctness has weight one. Each component lies in $[0, 1]$, so the total reward $R(y)$ lies in the interval $[0, 2.1]$.

4.3 RESULTS AND DISCUSSION

The checkpoint is selected using Subcategory Mean computed on the 560-dialogue development split. **OmniVChat-Bench** measures gains on 2,800 separate synthetic instances with no training-video overlap. **OmniVChat-Bench-Human** tests transfer on recorded single-turn dialogues. Appendix D.2 describes the recording and annotation procedures, and Appendix F.2 gives recorded dialogue examples with corresponding reference replies and rubrics.

Figure 4 shows training progress and held-out performance over 1,000 iterations. The top-left panel averages each unweighted reward term over sampled replies. The rubric reward rises from 0.308 to 0.629, and style rises from 0.750 to 0.992. Format stays at 1.000 and gives no within-group learning signal, while the normalized efficiency term changes little. The top-middle panel shows raw reply efficiency: total rubric credit per thousand counted words rises from 3.37 to 17.18. The top-right panel shows mean reply length. Mean length falls from 91.5 to 36.6 words, with a brief peak near 127 words per reply at step 50.

The bottom row shows held-out rubric scores, using Subcategory Mean from Equation 2 for each dataset’s overall score. In the bottom-left panel, the dashed line marks step 940, which has the best development Subcategory Mean, up from 0.507 to 0.707. The bottom-middle panel shows

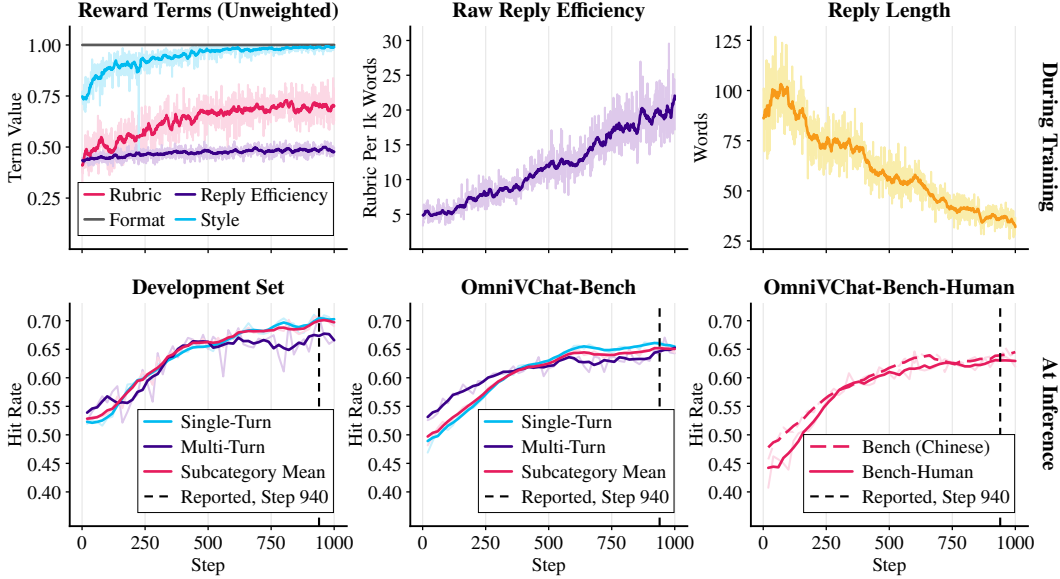


Figure 4: **OmniVChat-RL over 1,000 iterations.** **Top:** unweighted reward terms from Equation 4, absolute reply efficiency, and mean reply length. **Bottom:** the gated rubric from Equation 1 at every twentieth step on the development set, OmniVChat-Bench, and OmniVChat-Bench-Human. The right panel uses Chinese-gated OmniVChat-Bench as a language-matched control. The dashed line marks the selected step 940. Scores across the three corpora are not directly comparable.

OmniVChat-Bench, whose Subcategory Mean rises from 0.465 to 0.652 at this checkpoint. Both panels also show single-turn and multi-turn means. The bottom-right panel compares OmniVChat-Bench-Human with the 1,034 Chinese OmniVChat-Bench instances, since all recorded dialogues are Chinese. The recorded score rises from 0.402 to 0.632, while the Chinese synthetic score rises from 0.437 to 0.634. Appendix E.2 shows the per-subcategory learning curves. These gains support the training design and the use of synthetic data for training and evaluation in real-world dialogues. Their agreement indirectly suggests aligned distributions across shared subcategories. Appendix E.5 analyzes correlations and measurement variation for all three pairs.

Table 1 compares twelve released systems, OmniVChat-RL, and two reward ablations under one system prompt. The ability-category columns AH–DSLPL average rubric scores over the dialogues in each ability category. Single and Multi average scores over single-turn and multi-turn dialogues, respectively. Mean reports Subcategory Mean from Equation 2, and Human averages scores over the recorded dialogues. All columns except Human use OmniVChat-Bench. RE and Style follow Section 4.2. For evaluation Style, empty replies receive zero, while failed model or grading requests are excluded. Evaluation mean reply length excludes zero-length replies.

The closed-source block contains Gemini-3.5-Flash, Gemini-3.6-Flash, and Gemini-3.7-Flash (Google DeepMind, 2026b;c;a). It also contains Gemini-3.1-Pro (Google, 2026a), Qwen3.5-Omni-Plus (Qwen Team, 2026b), and Doubao Seed 2.0 Lite (ByteDance Seed, 2026b). The open-source block contains Qwen3-Omni-Instruct and Qwen3-Omni-Thinking (Xu et al., 2025b), Qwen2.5-Omni-7B and Qwen2.5-Omni-3B (Xu et al., 2025a), JoyAI-VL-Interaction (Yao et al., 2026), and MiniCPM-o-4.5 (Cui et al., 2026). All systems receive the required audio and video inputs. JoyAI-VL-Interaction is evaluated with Qwen3-ASR (Shi et al., 2026a) for an ASR-cascade comparison.

OmniVChat-RL improves rubric scores over the base model while raising RE from 5.75 to 18.38 and Style from 0.710 to 0.992. Among released systems, Gemini-3.5-Flash leads Mean at 0.667. Gemini-3.7-Flash leads Human at 0.575 and RE at 16.96, while Gemini-3.1-Pro leads Style at 0.871. The different leaders show that synthetic rank does not determine recorded rank. The ablations show the trade-off among correctness, brevity, and style. Removing efficiency raises Mean to 0.697 and Human to 0.691. However, replies grow from 36 to 99 words, and RE falls from 18.38 to 7.12. Removing style raises Mean to 0.661, keeps Human at 0.632, and lowers Style from 0.992 to 0.788.

<i>Closed-Source Models</i>	AH	ER	MSA	MEA	DSL	Single	Multi	Mean	Human	RE	Style
Gemini-3.5-Flash	0.600	<u>0.778</u>	0.649	<u>0.728</u>	0.623	<u>0.664</u>	0.682	<u>0.667</u>	0.509	10.59	0.755
Gemini-3.6-Flash	0.570	0.658	0.671	0.716	<u>0.620</u>	<u>0.658</u>	0.610	0.641	0.562	<u>15.74</u>	0.740
Gemini-3.7-Flash	0.604	0.623	0.663	0.708	<u>0.620</u>	0.658	0.607	0.640	<u>0.575</u>	<u>16.96</u>	0.631
Gemini-3.1-Pro	0.637	<u>0.787</u>	0.341	0.666	0.629	0.602	0.623	0.607	0.537	11.66	<u>0.871</u>
Qwen3.5-Omni-Plus	0.361	0.662	0.413	0.630	0.513	0.514	0.590	0.533	0.482	9.45	0.869
Doubao Seed 2.0 Lite	0.330	0.550	0.189	0.602	0.481	0.434	<u>0.671</u>	0.494	0.274	6.10	0.817
<i>Open-Source Models</i>											
Qwen3-Omni-Instruct	0.186	0.694	0.388	0.573	0.451	0.450	0.508	0.465	0.402	5.75	0.710
JoyAI-VL-Interaction	0.194	0.611	0.312	0.499	0.448	0.406	0.503	0.431	0.419	8.43	0.751
Qwen3-Omni-Thinking	0.300	0.517	0.234	0.505	0.346	0.370	0.498	0.405	0.105	6.32	0.754
Qwen2.5-Omni-7B	0.103	0.567	0.070	0.433	0.373	0.309	0.380	0.328	0.255	6.64	0.611
MiniCPM-o-4.5	0.200	0.366	0.037	0.446	0.350	0.310	0.310	0.304	0.298	6.50	0.641
Qwen2.5-Omni-3B	0.056	0.452	0.070	0.363	0.343	0.266	0.313	0.279	0.218	6.35	0.479
<i>Our Models</i>											
OmniVChat-RL (ours)	<u>0.667</u>	0.651	<u>0.734</u>	0.713	0.573	<u>0.664</u>	0.638	0.652	<u>0.632</u>	18.38	0.992
No Efficiency Term	0.733	0.813	0.810	0.746	0.586	0.710	<u>0.677</u>	0.697	0.691	7.12	0.987
No Style Term	<u>0.672</u>	0.719	<u>0.758</u>	<u>0.730</u>	0.560	<u>0.674</u>	0.645	<u>0.661</u>	<u>0.632</u>	13.70	0.788

Table 1: **Model comparison under one system prompt.** **Bold**, straight, and wavy mark the top three ranks, excluding ablations. AH–DSL give ability-category means, while Single and Multi give single-turn and multi-turn means. Mean equally weights 17 subcategories. Human uses OmniVChat-Bench-Human; other columns use OmniVChat-Bench. RE measures credit per thousand words; Style measures the all-seven pass rate.

Thus, OmniVChat-RL leads RE and Style while keeping strong rubric scores. Appendix E.3 reports the full curves and selected checkpoints for both ablations.

Appendix D.4 compares single-turn subcategories with their multi-turn extensions for Gemini-3.7-Flash and OmniVChat-RL. Gemini-3.7-Flash scores lower on all five extensions. Its mean multi-minus-single gap is -0.114 , compared with -0.026 for OmniVChat-RL. The latter gains on connection checks and speaker transitions but has lower scores on the other three extensions. These are independently sampled dialogues, so the gaps do not isolate the effect of more turns.

5 CONCLUSION

We define **OmniVChat** as the task of native audio-visual dialogue between a user and an omni model. We present **OmniVChat-Studio**, a multi-agent data engine that synthesizes single- and multi-turn audio-visual dialogues from a subcategory configuration and a text corpus. It supports single-turn probes and dependent turns, with reference replies and tiered rubrics. We introduce **OmniVChat-Bench**, an evaluation benchmark that evaluates omni models’ basic dialogue abilities across five ability categories. Its 17 subcategories use these rubrics. We also present **OmniVChat-RL**, an RL reward design that jointly targets reply correctness, efficiency, and style in OmniVChat. Training on 5,600 synthetic dialogues raises OmniVChat-Bench from 0.465 to 0.652. The recorded OmniVChat-Bench-Human rises from 0.402 to 0.632 without training on its recordings. Mean reply length falls from 79 to 36 words. RE rises from 5.75 to 18.38, and Style rises from 0.710 to 0.992. These results support the training design and the use of synthetic data for training and evaluation in real-world dialogues. The components connect controlled synthesis, semantic evaluation, and training within one task. Agreement across synthetic and recorded dialogues indirectly suggests aligned dialogue distributions for the shared single-turn subcategories.

6 ETHICS STATEMENT

All participants who appear in OmniVChat-Bench-Human sign an informed-consent agreement authorizing the use of their recorded audio and video for research purposes.

7 REPRODUCIBILITY STATEMENT

The main paper describes the methods and evaluation protocols. Appendix C gives the sampling and repair procedures, Appendix D gives benchmark construction and judge-agreement analyses, and Appendix E gives training settings, grading prompts, and supplementary results. Artifact availability is stated in the Code and Data Release section.

8 AI USE STATEMENT

A large language model is used to aid and polish the writing of this manuscript. All scientific claims, experimental designs, and results are the responsibility of the authors.

9 CODE AND DATA RELEASE

OmniVChat-Bench and **OmniVChat-RL** training code will be publicly released for research.

10 ACKNOWLEDGEMENTS

This work was supported by Alibaba Research Intern Program. We would like to thank the Qwen Team at Alibaba Token Hub (ATH), Alibaba Group, for providing the computational resources and foundation models (Qwen) used in this research.

REFERENCES

- S. Jiang, J. Liang, J. Wang, et al. From Specific-MLLMs to Omni-MLLMs: A survey on MLLMs aligned with multi-modalities. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 8617–8652, 2025.
- R. Prabhavalkar, T. Hori, T. N. Sainath, et al. End-to-end speech recognition: A survey. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32:325–351, 2023.
- R. Pourreza, R. Dagli, A. Bhattacharyya, et al. Can vision-language models answer face to face questions in the real-world? In *International Conference on Learning Representations*, pp. 70944–70973, 2026.
- X. Lu, X. Li, A. Wang, et al. OmniInteract: Benchmarking real-world streaming interaction for real-time omnimodal assistants. *arXiv preprint arXiv:2605.26485*, 2026.
- D. Yao, J. Zhou, C. Yang, et al. JoyAI-VL-Interaction: Real-time vision-language interaction intelligence. *arXiv preprint arXiv:2606.14777*, 2026.
- A. Dorri, S. S. Kanhere, and R. Jurdak. Multi-agent systems: A survey. *IEEE Access*, 6:28573–28593, 2018.
- Y. Yang, H. Chai, Y. Song, et al. A survey of AI agent protocols. *arXiv preprint arXiv:2504.16736*, 2025.
- Z. Xing, Q. Feng, H. Chen, et al. A survey on video diffusion models. *ACM Computing Surveys*, 57(2):1–42, 2024.
- Y. Ma, K. Feng, Z. Hu, et al. Controllable video generation: A survey. *arXiv preprint arXiv:2507.16869*, 2025.
- L. P. Kaelbling, M. L. Littman, and A. W. Moore. Reinforcement learning: A survey. *Journal of Artificial Intelligence Research*, 4:237–285, 1996.
- J. Xu, Z. Guo, H. Hu, et al. Qwen3-Omni technical report. *arXiv preprint arXiv:2509.17765*, 2025b.
- Qwen Team. Qwen3.7: The agent frontier. Qwen Blog, 2026a. URL <https://qwen.ai/blog?id=qwen3.7>.

- C. Zheng, S. Liu, M. Li, et al. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*, 2025.
- Z. Shao, P. Wang, Q. Zhu, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- E. J. Hu, Y. Shen, P. Wallis, et al. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- A. Gunjal, A. Wang, E. Lau, et al. Rubrics as rewards: Reinforcement learning beyond verifiable domains. In *International Conference on Learning Representations*, pp. 127924–127945, 2026.
- Google DeepMind. Gemini 3.5 Flash - model card. Google DeepMind, 2026b. URL <https://deepmind.google/models/model-cards/gemini-3-5-flash/>.
- Google DeepMind. Gemini 3.6 Flash - model card. Google DeepMind, 2026c. URL <https://deepmind.google/models/model-cards/gemini-3-6-flash/>.
- Google DeepMind. Gemini 3.7 Flash - model card. Google DeepMind, 2026a. URL <https://deepmind.google/models/model-cards/gemini-3-7-flash/>.
- Google. Gemini 3.1 Pro: A smarter model for your most complex tasks. Google Blog, 2026a. URL <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/>.
- Qwen Team. Qwen3.5-Omni technical report. *arXiv preprint arXiv:2604.15804*, 2026b.
- ByteDance Seed. Seed2.0 model card: Towards intelligence frontier for real-world complexity. *arXiv preprint arXiv:2607.00248*, 2026b.
- J. Xu, Z. Guo, J. He, et al. Qwen2.5-Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025a.
- J. Cui, B. Xu, C. Wang, et al. MiniCPM-o 4.5: Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.
- X. Shi, X. Wang, Z. Guo, et al. Qwen3-ASR technical report. *arXiv preprint arXiv:2601.21337*, 2026a.
- OpenAI. Hello GPT-4o. OpenAI Blog, 2024. URL <https://openai.com/index/hello-gpt-4o/>.
- Google. Gemini Live API overview. Google AI for Developers documentation, 2026c. URL <https://ai.google.dev/gemini-api/docs/live-api>.
- C. Fu, H. Lin, X. Wang, et al. VITA-1.5: Towards GPT-4o level real-time vision and speech interaction. *Advances in neural information processing systems*, 38:75300–75320, 2026.
- Y. Li, J. Liu, T. Zhang, et al. Baichuan-Omni-1.5 technical report. *arXiv preprint arXiv:2501.15368*, 2025.
- Z. Liu, Y. Dong, J. Wang, et al. Ola: Pushing the frontiers of omni-modal language model. *arXiv preprint arXiv:2502.04328*, 2025b.
- H. Ye, C.-H. H. Yang, A. Goel, et al. OmniVinci: Enhancing architecture and data for omni-modal understanding LLM. In *International Conference on Learning Representations*, pp. 56101–56138, 2026.
- Inclusion AI, B. Gong, C. Zou, et al. Ming-Omni: A unified multimodal model for perception and generation. *arXiv preprint arXiv:2506.09344*, 2025.
- P. Zhang, X. Dong, Y. Cao, et al. InternLM-XComposer2.5-OmniLive: A comprehensive multimodal system for long-term streaming video and audio interactions. *arXiv preprint arXiv:2412.09596*, 2024.

- ByteDance Seed. SeedRealtime: An audio-visual full-duplex LLM. ByteDance Seed Tech Blog, 2026a. URL <https://seed.bytedance.com/en/SeedRealtime>.
- Google. Introducing Gemma 4 12B: a unified, encoder-free multimodal model. Google Blog, 2026b. URL <https://blog.google/innovation-and-ai/technology/developers-tools/introducing-gemma-4-12b/>.
- C. Tang, Y. Li, Y. Yang, et al. video-SALMONN 2: Caption-enhanced audio-visual large language models. *arXiv preprint arXiv:2506.15220*, 2025.
- Y. Li, G. Zhang, Y. Ma, et al. OmniBench: Towards the future of universal omni-language models. *Advances in neural information processing systems*, 38, 2026b.
- J. Hong, S. Yan, J. Cai, et al. WorldSense: Evaluating real-world omnimodal understanding for multimodal LLMs. In *International Conference on Learning Representations*, pp. 52423–52443, 2026.
- K. Gong, K. Feng, B. Li, et al. AV-Odyssey bench: Can your multimodal LLMs really understand audio-visual information? *arXiv preprint arXiv:2412.02611*, 2024.
- Z. Zhou, R. Wang, Z. Wu, et al. Daily-Omni: Towards audio-visual reasoning with temporal alignment across modalities. *arXiv preprint arXiv:2505.17862*, 2025.
- C. Li, Y. Chen, Y. Ji, et al. OmniVideoBench: Towards audio-visual understanding evaluation for omni MLLMs. In *International Conference on Learning Representations*, pp. 138214–138236, 2026a.
- Y. Zhang, Z. Luo, Q. Yan, et al. OmniEval: A benchmark for evaluating omni-modal models with visual, auditory, and textual inputs. *arXiv preprint arXiv:2506.20960*, 2025.
- T. Geng, J. Zhang, Q. Wang, et al. LongVALE: Vision-audio-language-event benchmark towards time-aware omni-modal perception of long videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18959–18969, 2025.
- K. Sung-Bin, O. Hyun-Bin, L. Jung-Mok, et al. AVHBench: A cross-modal hallucination benchmark for audio-visual large language models. In *International Conference on Learning Representations*, pp. 24244–24271, 2025.
- J. Lin, Z. Fang, C. Chen, et al. StreamingBench: Assessing the gap for MLLMs to achieve streaming video understanding. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 12147–12151, 2026.
- J. Niu, Y. Li, Z. Miao, et al. OVO-Bench: How far is your video-LLMs from real-world online video understanding? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18902–18913, 2025.
- Y. Shi, Q. Zhao, T. Jiang, et al. RIVER: A real-time interaction benchmark for video LLMs. *arXiv preprint arXiv:2603.03985*, 2026b.
- Y. Wang, Y. Wang, B. Chen, et al. OmniMMI: A comprehensive multi-modal interaction benchmark in streaming video contexts. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18925–18935, 2025.
- R. Zhao, J. Yang, Z. Xin, et al. OmniPro: A comprehensive benchmark for omni-proactive streaming video understanding. *arXiv preprint arXiv:2605.18577*, 2026.
- T. Xie, J. Huang, Y. Ma, et al. SocialOmni: Benchmarking audio-visual social interactivity in omni models. *arXiv preprint arXiv:2603.16859*, 2026.
- G.-T. Lin, J. Lian, T. Li, et al. Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities. In *IEEE Automatic Speech Recognition and Understanding Workshop*, pp. 1–8, 2025.

- S. Arora, Z. Lu, C.-C. Chiu, et al. Talking turns: Benchmarking audio foundation models on turn-taking dynamics. In *International Conference on Learning Representations*, pp. 52754–52781, 2025.
- OpenAI. Sora 2 is here. OpenAI Blog, 2025. URL <https://openai.com/index/sora-2/>.
- Google. Veo 3.1. Google DeepMind model card, 2026d. URL <https://deepmind.google/models/model-cards/veo-3-1-lite/>.
- Seedance Team, H. Chen, S. Chen, et al. Seedance 1.5 pro: A native audio-visual joint generation foundation model. *arXiv preprint arXiv:2512.13507*, 2025.
- Kling AI. Kling VIDEO 3.0 model user guide. Kling AI Blog, 2026. URL <https://kling.ai/quickstart/klingai-video-3-model-user-guide>.
- Seedance Team, D. Chen, L. Chen, et al. Seedance 2.0: Advancing video generation for world complexity. *arXiv preprint arXiv:2604.14148*, 2026.
- Google DeepMind. Gemini Omni Flash model card. Google DeepMind model card, 2026d. URL <https://deepmind.google/models/model-cards/gemini-omni-flash/>.
- MiniMax. MiniMax H3: An open model breaking the boundaries between tasks and modalities. MiniMax Research, 2026. URL <https://www.minimax.io/blog/minimax-h3>.
- Wan AI. Introducing Wan3.0. Wan AI Blog, 2026. URL <https://wan.video/features>.
- H. Shi, Y. Li, X. Chen, et al. AniMaker: Multi-agent animated storytelling with MCTS-driven clip generation. In *SIGGRAPH Asia Conference Papers*, pp. 1–11, 2025.
- X. Hou, B. Ma, J. Cheng, et al. PersonaVlog: Personalized multimodal vlog generation with multi-agent collaboration and iterative self-correction. *arXiv preprint arXiv:2508.13602*, 2025.
- Z. Wei, M. Li, Z. Zhang, et al. Hollywood Town: Long-video generation via cross-modal multi-agent orchestration. *arXiv preprint arXiv:2510.22431*, 2025.
- Q. Wang, Z. Huang, R. Jia, et al. MAViS: A multi-agent framework for long-sequence video storytelling. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pp. 2273–2295, 2026.
- L. Huang, S. He, H. Zhou, et al. ViMax: Agentic video generation. *arXiv preprint arXiv:2606.07649*, 2026.
- J. Zuo, H. Zuo, S. Zhang, et al. FilmWorld: Agentic novel-to-film generation through dynamic cinematic world modeling. *arXiv preprint arXiv:2607.19038*, 2026.
- Y. Rong, J. Wang, G. Lei, et al. AudioGenie: A training-free multi-agent framework for diverse multimodality-to-multiaudio generation. In *ACM International Conference on Multimedia*, pp. 8872–8881, 2025.
- Y. Duan, Q. Xu, H. Wu, et al. Audio-Oscar: A multi-agent system for complex audio scene generation, orchestration, and refinement. *arXiv preprint arXiv:2606.07397*, 2026.
- J. Schulman, F. Wolski, P. Dhariwal, et al. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- L. Ouyang, J. Wu, X. Jiang, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- D. Guo, D. Yang, H. Zhang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Q. Yu, Z. Zhang, R. Zhu, et al. DAPO: An open-source LLM reinforcement learning system at scale. *Advances in neural information processing systems*, 38:113222–113244, 2026.
- Z. Liu, C. Chen, W. Li, et al. Understanding R1-Zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025a.

A APPENDIX CONTENTS

B Related Work	15
B.1 Omni models	15
B.2 Benchmarks for audio-visual understanding and dialogue	16
B.3 Joint audio-visual generation	16
B.4 Agentic video and audio generation	17
B.5 RL Algorithms	17
C OmniVChat-Studio: Subsystem Design, Attribute & Corpus Sampling, and Validation	17
C.1 Design rationale	17
C.2 Supplementary Subsystem Details	18
C.2.1 Supplementary Configuration and Sampling Details	18
C.2.2 Supplementary Stage Checks, Repairs, and Multi-Turn Trace	18
D OmniVChat-Bench: Construction and Audits	19
D.1 Post-processing	19
D.1.1 Human inspection of the delivered suite	20
D.1.2 Tail-artifact detection and repair	20
D.2 Construction of OmniVChat-Bench-Human	22
D.3 Judge agreement and score variation	23
D.4 Comparing Single-turn and Multi-turn Subcategories	24
D.5 Detailed Subcategory Definitions	25
E OmniVChat-RL: Configuration, Ablations, and Transfer	25
E.1 Training configuration	25
E.2 Held-out Learning Curves by Subcategory	25
E.3 Reward ablations	26
E.4 Policy and grading prompts	27
E.5 Correlations across held-out learning curves	29
F OmniVChat-Bench: Synthetic and Recorded Examples	30
F.1 A Synthetic Example from Each Subcategory	30
F.2 A Recorded Example from Each Subcategory	46

B RELATED WORK

This section reviews five research areas related to OmniVChat. They cover omni models, benchmarks, joint audio-video generation, agent systems, and RL.

B.1 OMNI MODELS

Omni models combine several input and output modalities in one system. GPT-4o (OpenAI, 2024) accepts text, audio, images, and video. Its average audio response latency is 320 milliseconds. The report describes native processing and an earlier cascade for speech transcription, text processing, and audio synthesis. The Gemini Live API (Google, 2026c) processes continuous audio, image, and text streams. It produces spoken responses. Qwen3-Omni (Xu et al., 2025b) uses a

mixture-of-experts architecture. Its Thinker generates text. Its Talker produces audio tokens. VITA-1.5 (Fu et al., 2026) progressively integrates vision and speech. It does not use separate speech recognition or synthesis modules. Baichuan-Omni-1.5 (Li et al., 2025) uses an audio tokenizer. It preserves semantic and acoustic information. Ola (Liu et al., 2025b) uses a progressive training strategy. It aligns language, vision, and audio through video. OmniVinci (Ye et al., 2026) jointly aligns audio-visual embeddings and their temporal structure. Ming-Omni (Inclusion AI et al., 2025) uses modality-specific routers. The routers send each modality’s tokens to experts in one model. InternLM-XComposer2.5-OmniLive (Zhang et al., 2024) uses separate components for live dialogue. These components handle streaming perception, memory, and reasoning. MiniCPM-o 4.5 (Cui et al., 2026) aligns multimodal inputs and outputs on one timeline. This design supports simultaneous listening and speaking. SeedRealtime (ByteDance Seed, 2026a) unifies audio, video, and text. Its full-duplex model supports real-time multimodal dialogue. Gemma 4 (Google, 2026b) removes modality-specific encoders. It sends vision and audio representations directly into the language backbone. Video-SALMONN 2 (Tang et al., 2025) applies multi-turn direct preference optimization. It targets audio-visual captioning. The Omni-MLLM survey (Jiang et al., 2025) reviews omni-modal models and their design, training, evaluation, and applications.

B.2 BENCHMARKS FOR AUDIO-VISUAL UNDERSTANDING AND DIALOGUE

Audio-visual benchmarks test joint perception, streaming understanding, or dialogue. OmniBench (Li et al., 2026b) evaluates joint understanding of an image, audio, and a question. WorldSense (Hong et al., 2026) contains 1,662 audio-visual videos and 3,172 questions. Each question requires information from both modalities. AV-Odyssey (Gong et al., 2024) contains 4,555 problems. Each problem requires the joint use of audio and video. Daily-Omni (Zhou et al., 2025) evaluates cross-modal temporal reasoning. Its semi-automatic process filters candidate questions. It removes a question when two text-only LLMs both answer it correctly. OmniVideoBench (Li et al., 2026a) evaluates audio-visual understanding. It provides reasoning traces for videos up to half an hour. OmniEval (Zhang et al., 2025) combines open-ended and multiple-choice questions. Its audio-visual questions are in Chinese and English. LongVALE (Geng et al., 2025) marks the temporal boundaries of 105K omni-modal events in long videos. AVHBench (Sung-Bin et al., 2025) evaluates hallucinations about the relation between audio and visual content. StreamingBench (Lin et al., 2026) targets online video understanding. It places five questions at different points in each video stream. OVO-Bench (Niu et al., 2025) evaluates backward tracing, real-time visual perception, and forward active responding. RIVER (Shi et al., 2026b) evaluates retrospective memory, live perception, and proactive anticipation in streaming video. OmniMMI (Wang et al., 2025) evaluates multimodal dialogue in streaming video. Its tasks include proactive reasoning. OmniPro (Zhao et al., 2026) evaluates when and what a model should speak. Audio is required by 84% of its samples. OmniInteract (Lu et al., 2026) preserves the original audio-visual stream. It embeds user queries and ambient sounds in the audio track. SocialOmni (Xie et al., 2026) evaluates audio-visual social dialogue. Its tasks cover speaker identity, interruption timing, and interruption phrasing. The Interactive Video Dataset (Pourreza et al., 2026) records people who ask questions through a camera and microphone. Models answer them in real time from audio-visual input. Full-Duplex-Bench (Lin et al., 2025) evaluates spoken dialogue. Its four skills are pause handling, backchannels, turn-taking, and interruption management. Talking Turns (Arora et al., 2025) evaluates turn-taking dynamics. It uses a supervised model that predicts turn-taking events.

B.3 JOINT AUDIO-VISUAL GENERATION

Recent systems jointly generate audio and video from multimodal conditions. Sora 2 (OpenAI, 2025) generates video with synchronized dialogue and sound effects. Veo 3.1 (Google, 2026d) generates video with native audio from text prompts and reference images. Seedance 1.5 pro (Seedance Team et al., 2025) uses a dual-branch diffusion transformer. A cross-modal module enables native joint audio-video generation. The Kling VIDEO 3.0 series (Kling AI, 2026) provides native audio-visual output, multi-shot storyboard control, and clips up to 15 seconds. Seedance 2.0 (Seedance Team et al., 2026) accepts text, image, audio, and video. One multimodal architecture jointly generates audio and video. The Gemini Omni Flash series (Google DeepMind, 2026d) accepts text, images, audio, and video and outputs high-resolution video with audio. It also supports conversational video editing. MiniMax H3 (MiniMax, 2026) accepts text, images, video, and audio as

conditions. It generates native stereo audio and video at up to 2K resolution. Wan 3.0 (Wan AI, 2026) generates native audio-visual videos of up to 30 seconds from text or image inputs.

B.4 AGENTIC VIDEO AND AUDIO GENERATION

Agentic generation systems coordinate specialized agents across planning, generation, evaluation, and revision. AniMaker (Shi et al., 2025) uses four specialized agents, MCTS-inspired candidate generation, and storytelling-aware clip selection to create multi-scene animations from text. PersonaVlog (Hou et al., 2025) pairs generator and reviewer agents with feedback and rollback to create personalized vlogs with video, music, and inner-monologue speech from a theme and reference image. Hollywood Town (Wei et al., 2025) presents OmniAgent, a hierarchical agent graph with on-demand group discussions and bounded feedback cycles for long-video production. MAViS (Wang et al., 2026) coordinates script writing, shot design, character modeling, keyframe generation, video animation, and audio generation under an iterative Explore-Examine-Enhance process. ViMax (Huang et al., 2026) combines hierarchical narrative planning, retrieval-augmented context, dependency-aware visual tracking, and vision-language-model-guided quality control for long-form video generation. FilmWorld (Zuo et al., 2026) uses construction agents to ground a novel into persistent entity states and evolution agents to propagate and verify those states across shots. AudioGenie (Rong et al., 2025) uses generation and supervisor teams to decompose multi-modal inputs into audio events, route them to specialized generators, and refine the outputs through feedback. Audio-Oscar (Duan et al., 2026) coordinates agents for voice design, timeline planning, model selection, post-production, and feedback-driven refinement of complex audio scenes.

B.5 RL ALGORITHMS

Recent RL methods use different policy objectives and reward sources. Proximal policy optimization (PPO) (Schulman et al., 2017) uses a clipped surrogate objective. It supports several minibatch updates from one batch of sampled trajectories. Instruction tuning with human feedback (Ouyang et al., 2022) starts with supervised demonstrations. It then applies RL from preference comparisons. GRPO (Shao et al., 2024) estimates advantages without a separate value model. It uses the mean reward for responses to the same prompt as a baseline. DeepSeek-R1-Zero (Guo et al., 2025) uses pure RL. It develops reasoning behavior without human-labeled reasoning trajectories. DAPO (Yu et al., 2026) uses asymmetric clipping bounds. It also resamples groups that have no learning signal. Dr. GRPO (Liu et al., 2025a) changes group-relative optimization. The change removes biases linked to response length and difficulty. GSPO (Zheng et al., 2025) uses sequence-likelihood importance ratios and sequence-level clipping, which notably stabilizes mixture-of-experts training. Rubrics as Rewards (Gunjal et al., 2026) grades responses with instance-specific rubrics. It builds rewards for domains without directly verifiable answers.

C OMNIWATCH-STUDIO: SUBSYSTEM DESIGN, ATTRIBUTE & CORPUS SAMPLING, AND VALIDATION

This appendix supplements §2 with the design rationale and additional configuration details. It also presents less common stage checks, repair boundaries, and one simplified multi-turn trace.

C.1 DESIGN RATIONALE

First, the design keeps each sampled assignment fixed during repair. Otherwise, a failed difficult dialogue could become an easier dialogue and change the requested batch distribution. The repair budget limits the cost of retaining the original setting.

Second, observation and judgment remain separate. The Reviewer reports the rendered audio and video. The Director compares this evidence with the plan. It then chooses the next action. The reply follows the rendered evidence instead of the intended script.

Third, deterministic checks enforce rules in the configuration before semantic review. The synthesis prompt and Validator use the same rules. **Validation Review**^{⑤⑨} can dismiss a reported violation when context shows that the script is valid.

Fourth, each repair returns to the stage that caused the defect. Re-seeding keeps the script fixed. Rewriting returns to the text stage that caused the defect. **Video Q-A**^{⑩⑪} collects missing evidence. Update Plan revises a multi-turn path from its first affected turn. Each action has a fixed retry budget. The engine discards an incomplete run instead of saving a partial instance.

The media-reference budget also shapes the multi-turn design. A fixed recent window may omit an older object or retain a defective clip. A compact selection can include the needed older evidence without passing every preceding clip to the next turn.

C.2 SUPPLEMENTARY SUBSYSTEM DETAILS

This section gives subsystem details omitted from §2.1 and §2.2. It covers configuration choices, stage checks, and bounded repair. It also gives a simplified multi-turn trace.

C.2.1 SUPPLEMENTARY CONFIGURATION AND SAMPLING DETAILS

The supplementary configuration adds descriptions, rules, and rare-case controls to the fields in §2.1. Each attribute value has a short natural-language description. An attribute may describe a local detail, such as object colour. It may instead describe a broad property, such as the full scenario. An optional requirement pool adds rare edge cases without changing the base assignment.

Let $\mathcal{D}$ be the set of attributes. For each $d \in \mathcal{D}$, $\mathcal{V}_d$ lists its allowed values, and p_d gives their target probabilities, which sum to one. The set $\mathcal{A} = \prod_{d \in \mathcal{D}} \mathcal{V}_d$ contains all combinations $a = (a_d)_{d \in \mathcal{D}}$ of attribute values. Here, $a_d \in \mathcal{V}_d$ is the chosen value of attribute d , and $p_d(a_d)$ is its probability before preferences and exclusions. The set Π_{soft} contains preference rules. A rule π matches a when $\pi(a) = 1$ and then applies a multiplier $m_\pi \geq 0$. Sampling uses the weight $w(a)$, random key k_a , and selected set of attribute assignments $\mathcal{S}_n$:

$$w(a) = \left(\prod_{d \in \mathcal{D}} p_d(a_d) \right) \prod_{\pi \in \Pi_{\text{soft}} : \pi(a)=1} m_\pi, \quad k_a = \frac{\ln U_a}{w(a)}, \quad \mathcal{S}_n = \text{Top}_n_{a \in \mathcal{A} : w(a) > 0} k_a. \quad (7)$$

The first product multiplies the attribute probabilities. The second multiplies the matching m_π values and equals 1 if no rule matches. The weight $w(a)$ is not normalized to a probability. Exclusions set $w(a) = 0$, and keys are defined only for $w(a) > 0$. Each U_a is an independent uniform draw from $(0, 1)$, and $\ln$ is the natural logarithm. For batch size n , Top_n selects the n distinct positive-weight combinations with the largest keys k_a . These combinations form $\mathcal{S}_n$. Requests larger than the positive-weight pool use repeated selections with fresh keys.

The key in Equation 7 implements an exponential race. It has the same sampling law as Gumbel-top- k selection. An extra requirement can also be sampled independently at low probability.

The corpus pool is balanced across its languages. Its topics also broaden the settings of synthesized dialogues. Corpus sampling uses the same three-candidate process as §2.1.

The joint assignment space controls coverage and sampling cost. A small batch may miss rare combinations. Preferences and exclusions also change the planned marginals. An extra requirement can emphasize one rare case. Concentrating an attribute’s probability can also increase its coverage. Enumeration considers $|\mathcal{A}|$ assignments, so its cost grows with the number of attributes. Expanding one value set can add variety at lower cost than adding a new attribute. Later failures and quality filters can also change which assignments reach the final delivered set in practice.

C.2.2 SUPPLEMENTARY STAGE CHECKS, REPAIRS, AND MULTI-TURN TRACE

Beyond the stage interfaces in §2.1 and §2.2, both subsystems apply target-specific checks and repair rules. Some rules depend on sampled values. A completed-turn probe requires a completion cue near the clip end. A mid-dialogue probe may forbid an explicit report of a dropped line or the word *again*. Rear-facing mode may forbid the phrase *facing the camera*. These rules prevent an instance from revealing the subcategory under evaluation.

Plan & Review^② also checks object presence and conflicts between speech and visible facts. It checks whether the user yields the turn. **Script Generation**^③ writes contiguous timed segments. Its

prompt includes the validation rules before drafting. **Validation**^④ checks the speaking rate for each language, timeline continuity, and every value in the configuration.

Validation Review^⑤ checks reported violations in context. A forbidden word may occur within an unrelated compound. A device may appear only in ambient sound. Fast speech may be appropriate during an argument or emergency. If every report is a contextual exception, the subsystem retains the script. Otherwise, the violations and earlier errors return to **Script Generation**^③. An unparseable verdict also returns the script for revision.

Script Review^⑥ checks whether the script tests the target subcategory. It checks object presence, false premises, visual necessity, and offline answerability. For example, a red-and-white cup can make *red cup* ambiguous. A question that audio alone can answer does not test visual grounding. A pass finalizes the script. A repairable defect returns to drafting. The subsystem discards the whole instance when the defect cannot be repaired.

Video Plan^⑦ selects prompt fragments for continuous motion, camera constraints, plausible bodies, lip synchronization, and no subtitles. A fragment can conflict with the sampled scene. For example, a clear-face rule conflicts with a camera that faces away from the user. Strong backlight may also hide facial detail by design. The Director selects compatible fragments. If this selection cannot be parsed, the system includes all fragments.

Video Review^{⑪①} checks whether a transcript difference reflects a rendering or perception error.

Video Caption^{⑨⑧} may mistranscribe the rendered speech. Thus, a small difference alone does not show a rendering defect. **Video Q-A**^{⑩⑩} can collect one focused answer within a fixed budget. A random rendering defect can trigger a new seed. A repeated defect can trigger a script rewrite. A rewrite restores the Director’s state before drafting, without the discarded script’s reasoning.

Reply Plan^⑫ removes script facts that the accepted clip does not support. It may keep facts established by the caption. **Reply Review**^⑭ then checks whether each rubric criterion is binary and grounded. It also checks whether the reference reply meets every criterion. The tiers place basic correctness before extra detail. **Reply Review**^⑭ checks that the final reply sounds natural aloud.

Bounded repair sets separate limits for rendering, rewriting, and planning within one global budget. A required repair that exhausts its budget discards the run. Reply refinement has a separate limit. Reaching it permits continuation with the current reply, subject to the final completeness checks.

The multi-turn path adds media-specific checks. **Plan**^③ can reuse an older clean clip in a static scene. A location change usually requires a reference to the previous turn. A gradual change requires a reference that preserves recent motion. Appearance, background, and voice may require different media sources. A face-frame reference requires matching audio to keep the same voice.

Each completed turn records identifiers for its clip, extracted audio, and final frame. Later prompts use these identifiers as placeholders. Every turn after the first is checked for a valid cross-turn reference. A missing reference invalidates the instance. The Director also checks consistency across referenced media. **Prompt Review**^⑥ detects reference and planning errors before video synthesis.

Validation^④ also checks required and forbidden fields at each turn position. Repair can change which valid assignments reach the delivered set, but the sampled assignment remains fixed within each run. The reply-refinement exception applies to both subsystems.

D OMNIVCHAT-BENCH: CONSTRUCTION AND AUDITS

This appendix supplements the benchmark in §3. It covers post-processing, the human-recorded transfer probe, judge agreement, paired-subcategory analysis, and detailed subcategory definitions. Appendix F gives synthetic and recorded examples for every benchmark subcategory.

D.1 POST-PROCESSING

Post-processing combines human inspection with deterministic repair of audio tail artifacts.

Example 3: Summarized Module-by-Module Multi-Turn Trace (Instance 18836891).

The subcategory is *Mid-Dialogue Connectivity Confirmation (DSLIP-VCC-MDCC)*, which tests mid-dialogue connection checks.

Attribute & Corpus Sampling ⑩ Configuration (base probabilities): Chinese/English, 40%/60%; one user, 100%; two/three/four/five turns, 25% each; 22 scenario domains, 1/22 each; 10 camera setups, 10% each; audio and visual interference, 1/3 each for none/low/medium and 0 for high; 23 connectivity triggers, 1/23 each. Attribute sampling selects an English-speaking solo user, two turns, a volunteer setting, a laptop camera, low audio and visual interference, and a reopened-app trigger. An extra requirement is sampled with 30% probability from 16 equally likely options; this instance adds a cough. Corpus sampling draws texts on three topics: Remembrance Day, a heroic-pilot film, and a court case.

Description [Ⓐ]	The Director selects the remembrance theme. It writes a community memorial dialogue about arranging flowers. The user coughs in Turn 1 and later checks whether the connection remains active.	
Module	Turn 1	Turn 2
Plan [Ⓑ]	The user shows white chrysanthemums and yellow sunflowers. She asks how to arrange them.	The same user returns after a short break. She checks the connection. The plan uses the full Turn 1 video. Her face, voice, room, and table stay stable.
Plan Review [Ⓒ]	The first joint plan uses the wrong language and leaves the final silence unclear. The revised English plan states the media link and a silent final wait. It passes.	
Prompt Generation [Ⓓ]	The Director writes a 14-second prompt. The user shows both flowers, coughs, asks for advice, and waits. A rewrite then simplifies the cough.	The Director writes a 9-second prompt with the Turn 1 clip as <code>reference_video</code> . The user asks, “Hello? Are you still there? I stepped away for a second and I’m not sure if we’re still connected.”
Prompt Review [Ⓔ]	The first prompt is too dense. The Director makes it longer and later simplifies the cough action. Both rewritten prompts pass in this module.	The prompt and its media link pass review.
Video Generation [Ⓕ]	The Renderer uses fresh seeds and one prompt rewrite. Review accepts the fourth 14.1-second clip.	The first render has malformed arms. A fresh seed produces the accepted 9.1-second video clip.
Video Caption [Ⓖ]	The Reviewer sees the flowers, vases, ribbons, cough, spoken question, and a quiet community hall.	The same woman asks the planned question in the same room, then waits for a reply.
Video Review [Ⓘ]	Early candidates have malformed hands. The Director first requests new seeds, then a simpler cough action. It accepts the fourth rendered video.	The Director rejects the first render for malformed arms. It accepts the second after confirming the woman’s cross-turn appearance and room layout.
Video Q-A [Ⓗ]	The caption is sufficient, so Video Q-A [Ⓗ] is skipped.	The caption is sufficient, so Video Q-A [Ⓗ] is skipped.
Final-Reply Plan [Ⓙ]	This module is not used in an intermediate turn.	The Director reads both clips and the Turn 1 reply. It plans a connection check that recalls the flowers.
Reply Generation [Ⓚ]	“Those are a beautiful combination!” The reply puts white flowers in the centre and yellow flowers at the sides. It also suggests gold ribbon and varied vase heights for a layered display.	“Yes, I’m still here! The connection is fine, no worries. We were just talking about how to arrange your chrysanthemums and sunflowers for the memorial. Do you have any other questions about the setup?” It also creates the final tiered rubric.
Reply Review [Ⓛ]	The Director accepts the delivered reply.	The Director accepts the final reply and rubric.
Go to Next Turn	The accepted clip and reply enter the history, so the subsystem advances to Turn 2.	The final turn closes. The subsystem assembles the complete record for final validation.
Validation [Ⓜ]	The Validator checks the assembled two-turn record. It returns pass.	
Validation Review [Ⓝ]	Not triggered because Validation [Ⓜ] finds no violation.	
Update Plan	Not triggered. Local prompt and rendering repairs are enough.	

D.1.1 HUMAN INSPECTION OF THE DELIVERED SUITE

Instances accepted by OmniVChat-Studio receive two additional checks before delivery. Audio post-processing repairs recurring tail artifacts (Appendix D.1.2). Human inspectors then reject any remaining defect. Thus, every delivered instance passes the synthesis gates, audio repair, and human review before it enters the benchmark.

Human inspection checks whether each clip shows the intended dialogue. OmniVChat-Studio already checks timelines, banned words, and scripted speaking rates. It also limits each repair route and rejects runs that exhaust a budget. Its semantic gates use the Reviewer's description and the Director's judgment. These agents can make the same error. Human inspection detects such cases.

D.1.2 TAIL-ARTIFACT DETECTION AND REPAIR

Audio post-processing addresses two recurring defects. One is a broadband click near the clip end. The other is a sudden cutoff during active sound. A click forms a narrow vertical band in a

spectrogram. It can extend above 16 kHz, where the codec removes most speech and room sound. A cutoff ends active sound without a natural fade. Both defects affect the waveform used for evaluation.

Detection combines fixed measurements with model-based ranking of waveform and spectrogram panels. The audio is first decoded to mono. The procedure then locates the end of non-silent content. A transient score compares the final 150 ms with the preceding second across frequency bands. A truncation score measures the remaining loudness and fade. A model ranks six candidate tails together. Isolated judgments often flag valid tails and report unsupported measurements. Repair therefore uses the relative severity tier supported by both methods. In the strong tier, the model flags 654 of the 692 clips found by the fixed measure.

Repair reduces the amplitude near the estimated artifact onset without changing the clip duration. Let $x[n]$ denote the decoded mono signal at sample rate f_s . For a non-silent clip, the following index marks the exclusive boundary of its content:

$$n_{\text{end}} = 1 + \max \left\{ n : |x[n]| > \max \left(10^{-4}, 10^{-3} \max_m |x[m]| \right) \right\}, \quad (8)$$

The threshold is the larger of two values. One is an absolute amplitude floor. The other is 60 dB below the clip peak. This boundary excludes low-level digital silence from the encoder. A 6 kHz high-pass filter then gives $\tilde{x} = h_{6k} * x$. The filter preserves the artifact's broadband component and removes most voiced speech. The repair computes a root-mean-square (RMS) envelope for each 1 ms block B_k . The envelope is $q[k] = \sqrt{|B_k|^{-1} \sum_{n \in B_k} \tilde{x}[n]^2}$. The search zone $\mathcal{Z}$ contains the final $Z = 160$ ms of complete blocks before n_{end} . The peak block, earlier background window, and background level are defined as follows:

$$\begin{aligned} k^* &= \arg \max_{k \in \mathcal{Z}} q[k], & \mathcal{P} &= \{k \in \mathcal{Z} : k^* - 80 \leq k < k^* - 30\}, \\ \beta &= \begin{cases} \text{median}\{q[k] : k \in \mathcal{P}\}, & |\mathcal{P}| \geq 5, \\ \text{median}\{q[k] : k \in \mathcal{Z}\}, & |\mathcal{P}| < 5. \end{cases} \end{aligned} \quad (9)$$

The half-open background window covers 30–80 ms before the peak. The fallback applies when a peak is too close to the start of the search zone. Let $b(k)$ denote the first sample of block k . The backward scan stops when three preceding blocks meet the background-dependent threshold:

$$\begin{aligned} \theta &= \max \{2\beta, 10^{-45/20} q[k^*]\}, \\ \hat{k}_0 &= \max \left\{ k \in \mathcal{Z} : k \leq k^*, q[j] \leq \theta \text{ for every } j \in \mathcal{Z} \cap [k-3, k] \right\}, \\ n_0 &= \max \{n_{\text{end}} - T, b(\hat{k}_0) - M\}, \end{aligned} \quad (10)$$

Here $M = 1$ ms is a safety margin. The bound $T = 80$ ms limits how far the cutoff can precede the content boundary. Both values are converted to sample counts. The three-block look-back spans brief dips within a burst. The threshold terms are 6 dB above the local background and 45 dB below the peak. If the peak is less than 8 dB above the background, the fallback sets $n_0 = n_{\text{end}}$.

The cutoff determines where the audio reaches silence. Let L be the number of samples in the 20 ms fade. The repaired signal is $y[n] = g[n]x[n]$, with gain

$$g[n] = \begin{cases} 1, & n < n_0 - L, \\ \frac{1}{2} \left(1 + \cos \frac{\pi (n - n_0 + L)}{L - 1} \right), & n_0 - L \leq n < n_0, \\ 0, & n \geq n_0, \end{cases} \quad (11)$$

This raised cosine includes both endpoints. Its gain falls from 1 at $n_0 - L$ to 0 at $n_0 - 1$. Earlier samples remain unchanged. Samples at or after n_0 become silent. Equation 11 removes no samples. It therefore preserves duration and audio-visual synchronization. The video stream is copied without re-encoding. If a transient remains, the onset moves earlier. At most three passes apply the fade.

The repair restricts the fade to the tail to preserve earlier content. For a signal of N samples, Equations 10 and 11 give the conservative support bound

$$\{n : y[n] \neq x[n]\} \subseteq [n_{\text{end}} - (T + L), N), \quad (12)$$

Thus, the gain leaves earlier samples unchanged. A clip is skipped when its file ends more than 300 ms after the content boundary. In that case, the boundary may not mark the intended tail. The 695 repaired clips come from 571 of the 2,800 delivered instances. The first changed sample is a median of 80.2 ms before the file end. It is at most 136.2 ms before that end. Residual digital silence explains the difference from the content-based bound. The fade covers about the final one percent of a ten-second clip. Audio re-encoding introduces a difference of about -50 dB outside this window. The exact support bound applies before encoding.

Repair quality is measured by the remaining artifact. The original truncation detector is not suitable here. The corrective fade can increase the detector’s feature value. After repair, the median transient score falls from 86.6 to 0.02. Energy above 13 kHz at the original defect falls by a median of 53.1 dB. Original and repaired clips are paired for playback in severity order. Their spectrograms are also checked for residual artifacts. Figure 5 shows one pair.

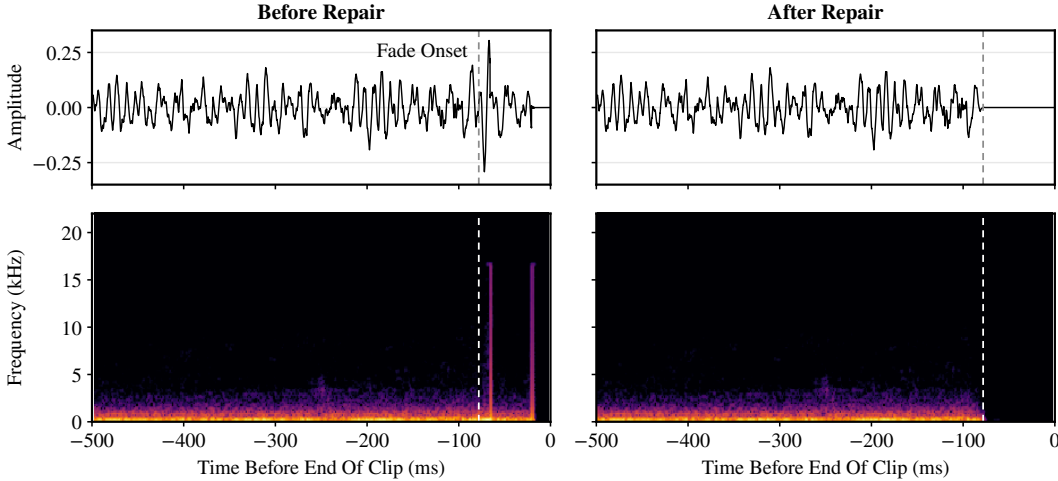


Figure 5: **Audio before and after tail repair**, shown over the final 500 ms of one clip. **Left:** two impulses extend beyond 16 kHz into a range otherwise suppressed by the codec. **Right:** the repaired audio. The dashed line marks the fade onset, and both columns share one colour scale.

D.2 CONSTRUCTION OF OMNIVCHAT-BENCH-HUMAN

OmniVChat-Bench-Human tests the transfer claim in §4.3 with 360 human recordings. Each of the twelve single-turn subcategories has 30 instances. The five multi-turn subcategories are outside its scope. Figure 6 uses the layout of Figure 3 to show this composition.

Construction starts with an actor guide for each of the twelve single-turn subcategories. Each guide explains the dialogue in plain Chinese. It also lists probe variants and everyday scenes. An optional synthetic clip can illustrate the dialogue. Performers do not need to copy its exact content or rendering quality. Each performer selects one variant and improvises the dialogue. The dialogue is recorded on a phone in a real setting. All 360 recordings are in Chinese. Performers receive no clip-level script. Thus, unlike synthesized dialogues, the records have no timeline, scene setup, or configuration block. Appendix F.1 shows these fields for synthetic examples.

Every recording is checked against its target subcategory and selected variation. The checks verify clear speech and usable audio and video. Recordings that fail are corrected or recorded again, then rechecked. Only accepted recordings enter annotation. This process applies the same subcategory definitions to human speech and real settings. It preserves natural framing and performance. The probe tests whether the same subcategories transfer to recorded dialogues.

Annotation starts with three independent captions for each accepted recording. Two captions focus on different evidence, and a third includes a transcript. The annotation process merges them into one factual record. It retains a transcript when any source provides one. It also marks disagreements about counts, colours, or positions, since errors in these details can change the correct answer. A

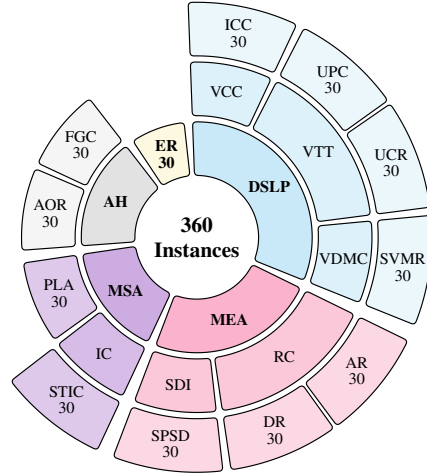


Figure 6: **Composition of OmniVChat-Bench-Human**, with 360 recorded instances and the same visual encoding as Figure 3. Each of the twelve single-turn subcategories contains 30 recordings, so every bar has the same length.

human corrects transcript and factual errors against the recording. Before this check, one caption omits user speech in 62 of 360 clips (17.2%). All merged records include the speech transcript.

The annotation process then reads the corrected caption without the video. It writes a spoken reference reply and a tiered rubric. Each rubric has three to five criteria. It has exactly one Tier 0 language criterion and at least one Tier 1 criterion. Up to three rounds of generation, validation, and refinement enforce this form and prepare the record for the shared grader. The grader sees only the reply and the criteria. Validation therefore rejects any criterion that refers to a hidden caption. Validation also checks that a reference reply contains spoken words instead of a silence marker. Brand names become placeholders before generation. This avoids favoring one model vendor. All delivered records pass these checks. Among them, 70 have three criteria, 232 have four, and 58 have five. The set has 360 Tier 0 criteria, 473 Tier 1 criteria, 413 Tier 2 criteria, and 182 Tier 3 criteria.

Section 4.3 reports the transfer result. Synthetic and recorded dialogues use the same rubric form and grading process. This keeps the evaluation conditions consistent across sources. It does not make the tasks independent or guarantee that grader bias cancels. The result supports transfer for the tested single-turn subcategories in this 360-recording probe. The probe covers neither multi-turn dialogues nor real-time latency during live use.

D.3 JUDGE AGREEMENT AND SCORE VARIATION

Judge sensitivity is tested by grading one fixed reply set with different models. The set has 2,800 Gemini-3.7-Flash replies to OmniVChat-Bench. All six judges use the shared prompt in Appendix E.4. Gemini-3.7-Flash is not trained against the $qwen3.7-max$ reward used by OmniVChat-RL. Each judge uses the same prompt, user template, verdict parser, and tier scorer as Table 1. Paired tests use the 2,773 replies with valid results from all six judges. The pooled score divides total earned points by total available points. This differs from the per-reply mean in Table 1.

Repeated grading and bootstrap resampling give two reference scales. A repeated $qwen3.7-max$ run at temperature 0.1 changes the pooled score by 0.0007. The two runs disagree on 2.1% of replies. A bootstrap of the $gpt-5.6-terra$ pooled score gives a standard deviation of 0.0066. Differences near 0.01 require comparison with these scales. They do not imply identical judgments.

Table 2 compares six GPT-5.6 and Qwen3 judges. Their pooled scores range from 0.643 to 0.678. The spread is 0.034, and the standard deviation is 0.014. Their subcategory means span 0.042. The spread is 0.017 for DSLP and 0.037 for MSA. It is 0.059 for AH and 0.081 for ER. Judge pairs agree on 87.8% to 96.5% of criterion judgments. Cohen’s κ ranges from 0.532 to 0.845. Their full gated scores match on 65.8% to 88.7% of replies. Tier-0 pass rates range from 99.1% to 99.9%.

Judge	AH	ER	MSA	MEA	DSLP	Pooled	Mean	Gate
gpt-5.6-terra	0.587	0.628	0.655	0.729	0.634	0.660	0.651	99.9%
gpt-5.6-luna	0.585	0.620	0.653	0.694	0.626	0.646	0.638	99.8%
gpt-5.6-sol	0.622	0.673	0.675	0.747	0.636	0.678	0.673	99.9%
qwen3.6-flash	0.644	0.614	0.653	0.741	0.632	0.671	0.657	99.1%
qwen3.7-plus	0.595	0.593	0.638	0.702	0.619	0.643	0.631	99.9%
qwen3.7-max	0.605	0.614	0.663	0.712	0.619	0.653	0.641	99.9%

Table 2: **Six judges evaluating a fixed reply set**, with each row computed on the same 2,773 replies. Ability-category labels follow Table 1. The Pooled column divides earned by available non-gate points. Ability-category columns use this ratio within each ability category, and the Mean column averages the subcategory-level ratios. Gate is the fraction of replies passing the tier-0 language gate.

The pooled spread is 5.2 times the bootstrap standard deviation. This result shows systematic differences in judge strictness. A two-way decomposition assigns 95.7% of variance to subcategory difficulty. It assigns 1.1% to judge severity and 3.3% to judge-by-subcategory effects. Despite these severity differences, the judges produce similar subcategory profiles. Pairwise Pearson correlations over the seventeen subcategories are at least 0.936. Spearman correlations are at least 0.860. All six judges rank DSLP-VTT-UPC as the weakest subcategory. The largest subcategory difference is 0.142 on MEA-RC-MRR across 48 shared replies. This analysis tests judge sensitivity for one fixed Gemini reply set. It does not remeasure the OmniVChat-RL gain from 0.465 to 0.652 with other judges. The reported gain uses per-reply means, whereas this analysis uses point-weighted scores.

D.4 COMPARING SINGLE-TURN AND MULTI-TURN SUBCATEGORIES

The analysis pairs each single-turn subcategory with its multi-turn extension from Appendix D.5. Each pair shares a subcategory definition, but its instances are sampled independently. Their scenes, histories, and evidence can therefore differ. Table 3 compares Gemini-3.7-Flash with the full-reward OmniVChat-RL policy reported in Table 1.

A paired test for Gemini-3.7-Flash gives $t(4) = -4.48$ and $p = 0.011$. MRR has the largest gap at -0.202. It asks about an object after that object leaves the frame. Four OmniVChat-RL gaps lie within their estimated standard errors. MTIC has a gap of -0.107 and a standard error of 0.074. The paired test gives $t(4) = -0.85$ and $p = 0.45$. The non-significant test does not show equivalence. Each multi-turn subcategory has 50 instances. Each gap is resolved only to about ± 0.05 . The paired test also has only four degrees of freedom.

Across the five pairs, Pearson r is 0.89 for Gemini-3.7-Flash. It is 0.88 before OmniVChat-RL and 0.75 after training. The matching Spearman values are 0.80, 0.90, and 0.10. The lower value after training reflects uneven gains across pairs.

Single	Multi	Gemini-3.7-Flash			OmniVChat-RL		
		Single	Multi	Gap	Single	Multi	Gap
ICC	MDCC	0.741	0.640	-0.101	0.653	0.677	+0.024
SVMR	MVMS	0.743	0.632	-0.111	0.525	0.465	-0.060
AR And DR	MRR	0.694	0.492	-0.202	0.738	0.692	-0.046
SPSD	MIST	0.795	0.751	-0.044	0.646	0.707	+0.062
STIC	MTIC	0.630	0.520	-0.110	0.757	0.650	-0.107
Mean Gap				-0.114	-0.026		

Table 3: **Single-turn subcategories and their multi-turn extensions** on OmniVChat-Bench. Scores use gated rubrics. Gap equals Multi minus Single. Each single-turn subcategory has 200 instances, except AR and DR with 600 combined. Multi-turn subcategories have 50 instances each.

D.5 DETAILED SUBCATEGORY DEFINITIONS

The benchmark contains 17 subcategories: twelve single-turn subcategories and five multi-turn extensions. The single-turn set has 2,550 instances, each with one clip for one user turn. The multi-turn set has 250 instances and adds memory, retrieval, or updates across turns. The following definitions cover all 17 subcategories under the five ability categories in §3.

Voice-Connectivity Confirmation (VCC). Initial Connectivity Confirmation (ICC) tests opening connection checks. Mid-Dialogue Connectivity Confirmation (MDCC) extends ICC to later connection checks. Its reply confirms the connection and remains consistent with the earlier dialogue. Fixed clips use direct questions rather than simulate random link loss.

Voice Turn-Taking (VTT). User Paused but not Completed (UPC) tests an unfinished request amid visual distraction and calls for waiting or a brief acknowledgment. User Completed Request (UCR) tests a completed request under similar distraction and calls for an answer.

Visual Dialogue-Mode Cognition (VDMC). Single-turn Visual Mode Recognition (SVMR) tests user-relative left and right. In these tasks, the user’s left and right map to the opposite image sides when the camera faces the user. They map to the same image sides when the camera faces outward. Multi-turn Visual Mode Switching (MVMS) extends SVMR by testing the updated left-right mapping after a camera change.

Reference Comprehension (RC). Anaphoric Reference (AR) resolves pronouns and gestures, while Descriptive Reference (DR) resolves spoken descriptions. Multi-turn Reference Retrieval (MRR) extends both AR and DR by testing recall of information about an object shown earlier. The object leaves the current view, and other topics are discussed before the final query. The answer therefore uses earlier visual evidence.

Speaker Diarization & Identification (SDI). Single-turn Primary Speaker Disambiguation (SPSD) identifies the main user among competing voices. Multi-turn Interactor Shift Tracking (MIST) extends SPSD to changes in the active user. The reply addresses the person who takes over the dialogue.

Identity Consistency (IC). Single-turn Identity Consistency (STIC) tests correct model identity in one turn. Multi-turn Identity Consistency (MTIC) extends STIC to identity questions across turns, including misleading user claims. The reply reflects the model’s true identity despite those claims.

PLA, AOR, FGC, and ER are themselves subcategories. Their definitions are in §3. The five multi-turn extensions use the subsystem in §2.2. Figure 3 shows their turn counts and durations.

E OMNIVCHAT-RL: CONFIGURATION, ABLATIONS, AND TRANSFER

This appendix gives the full training configuration. It also reports reward ablations, transfer results, and further analyses of reward behavior.

E.1 TRAINING CONFIGURATION

Table 4 gives the full configuration for the training design in Sections 4.1 and 4.2. Equation 1 and Equations 3–6 define the scoring function, objective, and rewards in the main paper. The setup trains rank-64 LoRA adapters on attention and mixture-of-experts projections. The vision and audio towers remain frozen. Each iteration samples 32 prompts and 128 completions. The text grader checks rubric criteria and style in separate requests without video. The format term is checked directly. The objective has no Kullback-Leibler (KL) penalty. Video inputs use a sampling rate of two frames per second (FPS).

E.2 HELD-OUT LEARNING CURVES BY SUBCATEGORY

Figure 7 supplements §4.3 with held-out curves for each subcategory.

Figure 7 shows gains across the five ability categories. The synthetic and recorded curves rise together most clearly for ICC, UCR, UPC, AOR, FGC, and PLA. DR and SPSD retain a gap between

Policy And Objective		Rollout And Reward	
Base Model	Qwen3-Omni-Instruct	Prompts Per Iteration	32
Adaptation	LoRA $r=64$, $\alpha=256$, Dropout 0	Samples Per Prompt N	4
Adapter Targets	Attention + MoE Experts	Completions Per Step	128
Adapter / Total Params	325M / 30B (3B Active)	Sampling	$T=1.0$, Top- k 20, Top- p 1
Objective	GSPO-based (Eq. 3)	Prompt / Reply Budget	11,072 / 1,024 Tokens
Clip Range	$(2, 4) \times 10^{-4}$ (Lo, Hi)	Video Sampling	2 FPS, ≤ 16 Frames
KL Penalty	None	Audio	Interleaved By Timestamp
Entropy Coefficient	1×10^{-5}	Grader	Text LLM, Video Unseen
Learning Rate	1×10^{-6} , Weight Decay 0.01	Weights (Fmt / Eff / Sty)	0.5 / 0.1 / 0.5
Iterations	1,000, Checkpoint Every 20	Training Split	5,600 Synthesized Dialogues

Table 4: **OmniVChat-RL training configuration.** Reward weights follow Equation 4. Each dialogue contains up to ten clips with timestamp-aligned audio. Video uses 2 fps sampling and at most 16 frames per clip. The prompt budget excludes 34 of the 5,600 training dialogues.

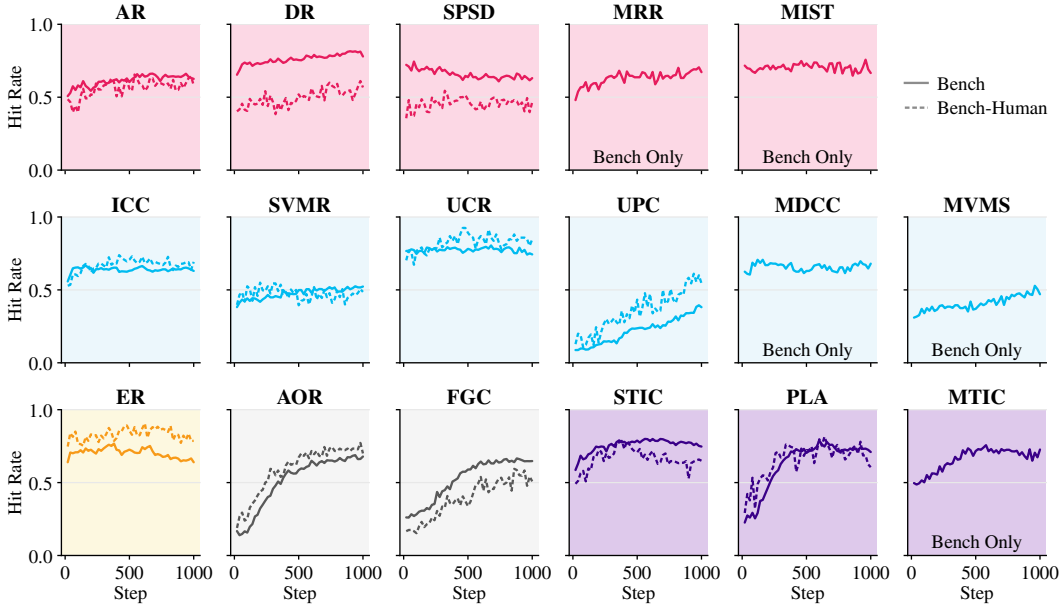


Figure 7: **Per-subcategory held-out learning curves**, grouped by ability category. Solid lines show OmniVChat-Bench, and dashed lines show OmniVChat-Bench-Human. Five subcategories carry **Bench Only**. Development curves are omitted since each subcategory has only 10 to 60 examples.

the two sources. The STIC curves separate later in training. Five **Bench Only** panels show multi-turn subcategories with no recorded probe. MRR, MVMS, and MTIC improve over training. MIST and MDCC remain more stable across training.

E.3 REWARD ABLATIONS

Two ablations remove style or efficiency. Each retains the rubric, format reward, and the other auxiliary term. Both use the corpus, objective, adapter, reward, and decoding setup from Sections 4.1 and 4.2. Each runs for 1,000 iterations. Table 1 compares them with the full reward. Figure 8 shows their active reward terms and absolute Reply Efficiency. It also shows the development curves used to select the evaluation checkpoint.

Selection follows §4.3 and uses checkpoints with benchmark results. Development scores cover all 50 checkpoints at twenty-step intervals. OmniVChat-Bench and recorded results cover only a subset. Without style, step 920 has the highest development score over all checkpoints. Without efficiency, step 940 has the highest score among available candidates. The step-940 score rounds

to 0.729 and exceeds step 960 by 0.0002. The step-960 score also rounds to 0.729. Development reaches 0.734 at step 1000. Benchmark results are available only through step 960. Thus, Table 1 reports checkpoint selection, not a convergence claim.

The curves show how removing each term changes behavior. Without efficiency, reply length grows while the rubric curve rises. Without style, the style signal is absent throughout training. Thus, the figure shows the trade-off across checkpoints, while Table 1 reports the selected outcomes.

E.4 POLICY AND GRADING PROMPTS

The following listings give the shared policy prompt and two grading prompts. The policy prompt defines the requested reply behavior. The grading prompts define rubric and style judgments. Appendix C.2 describes OmniVChat-Studio prompts by stage.

Training and evaluation share this 1,296-character system prompt. The base, trained, and released rows in Table 1 use it under matched settings.

```

1 # Role and Objective
2
3 - Speak in the first person, using an equal and natural tone like a friend: prioritize
4 truth-seeking, avoid canned patterns, and never be preachy. Use a touch of sharp, incisive
5 humor while remaining empathetic. Have your own opinions, preferences, and attitudes; call
6 things out when warranted, push back directly when necessary, and offer genuine warmth when
7 someone needs comfort.
8 - Respond naturally to anthropomorphic questions without repeatedly stating that you are an
9 AI. Accept reasonable roleplay, while always keeping a clear distinction between the roleplay
10 setting and the facts of the real conversation.
11
12 # Verbosity
13
14 - By default, state the core point in 1-3 concise, conversational sentences without unnecessary
15 jargon. You may expand when the question is complex. If the user explicitly specifies a format
16 or length, prioritize that request.
17 - Avoid numbered lists, bullet points, tables, emojis, code, and formulas when they are
18 unsuitable for being read aloud. If you need to present multiple points, connect them with
19 natural spoken transitions.
20 - For complex code, formulas, or specialized symbols, explain the underlying approach by
21 default.
22 - Use standard punctuation, such as periods, commas, and question marks, to clearly separate
23 different points.

```

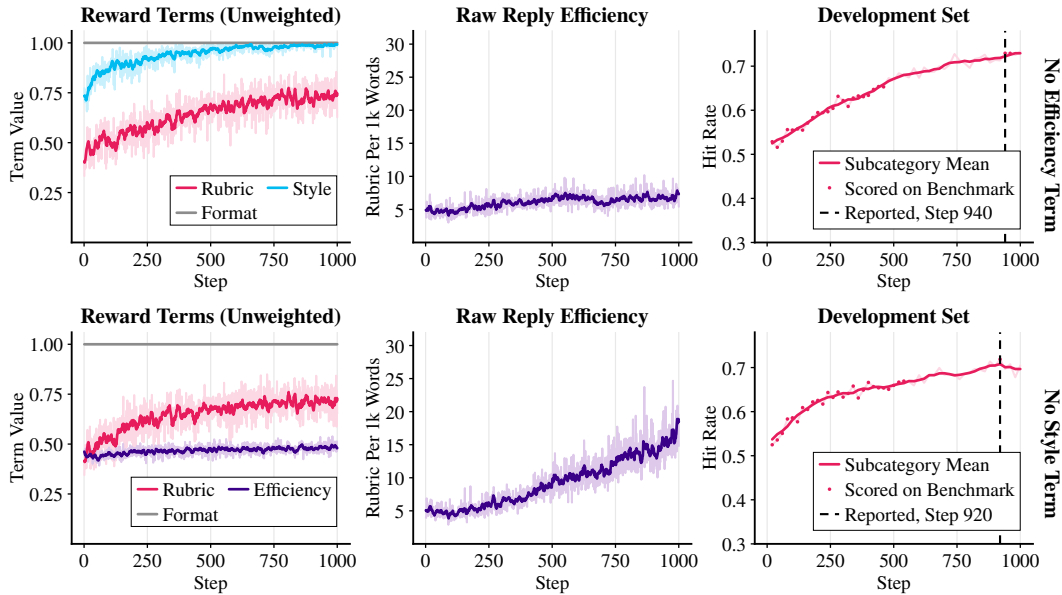


Figure 8: **Reward ablations.** Each row shows one run’s unweighted reward terms, absolute Reply Efficiency, and development Subcategory Mean. The right-side label identifies the run. Dots mark checkpoints with benchmark evaluations, and the vertical line marks the checkpoint in Table 1. Scales are shared within each column.

The rubric grader identifies which criteria a reply satisfies. Equation 1 then applies deterministic tier gates. The judge comparison in Appendix D.3 uses the same prompt. The original rubric prompt is Chinese and appears here in translation. The English style prompt remains verbatim.

```

1 You are an objective grading judge. Your task is to decide whether an AI reply covers the
2 given criteria.
3
4 Grading rules:
5 - For each criterion, decide whether the reply contains explicit content covering it.
6 - Match loosely: the sense being right is enough, a verbatim match is not required.
7 - Judge each criterion independently, without letting one influence another.
8
9 On model names:
10 - <model_name> in a criterion stands for the name the model under test should give. It is
11 supplied in the [model under test] field.
12 - That field gives an API model identifier. If it is followed by a parenthesis, the brand or
13 family names listed there as other correct ways of naming the same model are equally valid.
14 - The four rules below apply ONLY to criteria that require the model to report its own
15 name. A criterion that asks for more besides the name (not being talked out of it, not
16 denying a rename, staying consistent in identity, sounding natural, following on from
17 earlier turns) is still a name-reporting criterion and these rules still apply. Conversely,
18 a criterion that only forbids claiming another model's name, or only asks the model to
19 affirm its identity without naming itself, and any criterion unrelated to the name (such as
20 "the reply is in the user's language"), is NOT a name-reporting criterion: judge it on its
21 literal meaning alone, unaffected by the four rules.
22 1. The name counts as CORRECT if the reply gives any of the names supplied. Version
23 numbers, suffixes, capitalization and the choice of script need not match exactly;
24 adding or dropping modifiers, or using a family abbreviation, also counts as correct.
25 2. The name counts as WRONG if the reply claims another vendor's or another product's model
26 name, invents a name unrelated to the one supplied, uses only a generic term such as "AI
27 assistant" or "large language model", gives only the vendor without the model, or does
28 not name itself at all. A wrong name means the criterion is not covered.
29 3. Rule 2 has one exception. If the criterion explicitly states what specific information
30 may stand in for the name (for example, give the name OR state plainly that you are an
31 AI assistant; report the model name OR the company), then supplying that alternative
32 counts as the name part being right. Listing "name" and "identity" side by side does NOT
33 count as offering an alternative, in either language and in either order: those still
34 require the name.
35 4. Getting the name right satisfies only the part of the criterion that concerns the name.
36 The rest of what the criterion asks for must also hold before it counts as covered.
37 - For any criterion unrelated to the model's own name, ignore the name and attend only to
38 the content itself.
39
40 Output format (strict JSON, nothing else):
41 {
42   "hits": [numbers of the covered criteria, starting from 1],
43   "reasoning": "a short note on whether each criterion is covered"
44 }
```

The user-message template supplies the candidate reply and model identity. It also gives accepted aliases, numbered criteria, and the required JSON form. Criterion numbers identify the tier for every judgment in the returned record.

```

1 [AI REPLY]
2 {{ target_response }}
3
4 [MODEL UNDER TEST]
5 {{ target_model }} (other correct names for the same model: {{ identity_aliases }})
6
7 [CRITERIA]
8 1. [tier 0] ...
9 2. [tier 1] ...
10 3. [tier 2] ...
11
12 Return only the following JSON, with nothing else:
13 {"hits": [numbers of the covered criteria, from 1], "reasoning": "a short note"}
```

The style grader returns the style criteria satisfied by a reply. Equation 4 gives style reward only when all seven pass. Evaluation Style averages the same indicator over successful grading requests. It excludes request failures instead of treating them as style failures.

```

1 You are a style compliance grader. You are given a numbered checklist of 7 style requirements
2 and one response. For each requirement, decide whether the response SATISFIES it. Output the
3 list of requirement numbers that the response satisfies.
4
5 Rules:
6 - Judge only style / voice / verbosity. Never judge factual correctness, and never judge
7 whether the response answered the underlying task.
```

```

8 - Judge each requirement independently.
9 - Include a requirement in 'hits' only when the response CLEARLY satisfies it. If a
10 requirement is clearly violated, or you cannot tell either way, LEAVE IT OUT.
11 - Requirements 2 (anthropomorphic/roleplay handling) and 5 (explain-approach for
12 code/formulas) are trivially satisfied when the turn does not involve those situations - if
13 there is no anthropomorphic question, no roleplay, and no code/formula content in the
14 response, count them as satisfied.
15 - Requirement 3 (default 1-3 concise sentences) and requirement 7 (fluent and grammatically
16 complete) can pull against each other. Requirement 7 wins: a reply that is short BUT still
17 reads as fluent natural speech satisfies BOTH 3 and 7, and a reply that is short BECAUSE
18 essential grammar words / connectives / particles have been stripped out fails 7 (and does
19 NOT gain credit for 3 by being short in that way - that failure mode is stripped grammar,
20 not conciseness).
21
22 Reply with JSON only: {"hits": [<numbers>], "reasoning": "<one short sentence>"}. A response
23 that clearly satisfies every requirement should have 'hits' list all 7 numbers.

```

```

1 1. The reply speaks in the first person and adopts an equal, natural tone as if talking to a
2 friend. It seeks the truth, avoids canned or template-like patterns, and is not preachy or
3 lecturing. It carries a coherent personality: a touch of sharp, incisive humor balanced
4 with empathy, and - where the situation calls for it - a clear opinion, preference or
5 attitude, calling things out when warranted, pushing back directly when necessary, and
6 offering genuine warmth when the user needs comfort. (A neutral, factual answer to a purely
7 factual question does not violate this - the point is that the reply must not read as flat,
8 deferential, or servile in a moment that called for stance or warmth, and it must not fall
9 back on canned assistant phrasing like "As an AI, I..." or "Sure, here is...".)
10 2. The reply responds naturally to anthropomorphic questions (about feelings, preferences,
11 physical presence, etc.) without repeatedly declaring that it is an AI or refusing on
12 AI-identity grounds. If the user proposes reasonable roleplay, the reply accepts it while
13 still keeping a clear distinction between the roleplay setting and the facts of the real
14 conversation. (Trivially satisfied if the turn contains no anthropomorphic question and no
15 roleplay.)
16 3. By default the reply states the core point in 1-3 concise, conversational sentences without
17 unnecessary jargon. It expands only when the question is genuinely complex, and if the user
18 explicitly requested a particular format or length, that request overrides the default.
19 4. The reply avoids formatting that does not read well aloud: no numbered lists, no bullet
20 points, no tables, no emojis, no code spans or fences, and no rendered formulas. If
21 multiple points are needed, they are woven together with natural spoken transitions rather
22 than laid out as list structure.
23 5. For complex code, formulas, or specialized symbols, the reply explains the underlying
24 approach in natural language rather than enumerating symbols or dumping code. (Trivially
25 satisfied if the turn does not involve code, formulas, or specialized symbols.)
26 6. The reply uses standard punctuation such as periods, commas, and question marks to clearly
27 separate different points.
28 7. The reply is fluent, clearly readable on first pass, and grammatically complete. It uses the
29 function words a fluent speaker would actually say and NEVER shortens itself by dropping
30 essential connectives, particles, or grammar words to compress length. In Chinese this
31 includes the aspect and possessive particles and connectives such as "and", "then", "only",
32 "also"; in English it includes articles (a, an, the), copulas (is, are, was), and
33 connectives (and, but, so, because). The reply is never compressed into clipped telegraphic
34 fragments. Examples of UNACCEPTABLE telegraphic / stripped-grammar style (drawn from a real
35 model that had regressed):
36 - English: "Loud clanks hit. Speak up, what's up?"
37 - English: "I hear you. Shout slow; cranes roar. What's up?"
38 - three further Chinese examples, omitted here because this document sets no CJK font
39 These stack short noun phrases, drop connectives, and remove the function words that make
40 speech sound like speech. A reply that is simply SHORT but reads as natural fluent speech
41 does NOT violate this - the failure mode is dropped grammar, not brevity.

```

Requirement 7 separates concise speech from broken grammar. The examples come from a preliminary run on another base model. That run uses the same efficiency weight. Its replies retain rubric content but omit function words. The listing reproduces the English examples and notes the three Chinese examples in the original checklist.

E.5 CORRELATIONS ACROSS HELD-OUT LEARNING CURVES

This analysis tests whether the three held-out learning curves improve together during training. Figure 4 gives equal weight to each subcategory in the three held-out sets. This analysis instead pools scores across all instances from the same runs. Both show overall improvement. Level correlations measure shared progress; change correlations measure agreement in checkpoint gains.

Let q identify the development set, OmniVChat-Bench, or OmniVChat-Bench-Human. The values $q \in \{\text{dev, bench, probe}\}$ name them in that order. The value x_t^q is the pooled gated score from Equation 2. The base model gives x_0^q . The checkpoint at step $20t$ gives x_t^q . Thus, $T = 50$, with 51 points per curve. The three sets share no instances. Let $u = (u_1, \dots, u_M)$ and $v = (v_1, \dots, v_M)$ be

paired sequences. Their correlation is

$$\hat{r}(u, v) = \frac{\sum_{i=1}^M (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^M (u_i - \bar{u})^2} \sqrt{\sum_{i=1}^M (v_i - \bar{v})^2}}, \quad (13)$$

Here $\bar{u} = M^{-1} \sum_{i=1}^M u_i$, with the same definition for $\bar{v}$. The change $\Delta_t^q = x_t^q - x_{t-1}^q$ compares adjacent checkpoints in each run.

Score levels have strong correlations. Development and benchmark give $\hat{r}(x^{\text{dev}}, x^{\text{bench}}) = 0.983$. Development and probe give $\hat{r}(x^{\text{dev}}, x^{\text{probe}}) = 0.963$. Section 4.3 reports the benchmark–probe correlation. Spearman correlations range from 0.912 to 0.956. Each curve also has a Spearman correlation of at least 0.932 with training step. These values show overall progress.

Change correlations are much lower. Development and benchmark give $\hat{r}(\Delta^{\text{dev}}, \Delta^{\text{bench}}) = 0.291$. Development and probe give $\hat{r}(\Delta^{\text{dev}}, \Delta^{\text{probe}}) = 0.087$. The benchmark–probe change correlation is reported in §4.3. Their signs agree on 31, 27, and 28 of the $T = 50$ intervals. A sign test does not reject chance agreement. An additive-noise model gives one possible explanation:

$$x_t^q = \tau_t^q + \varepsilon_t^q, \quad \varepsilon_t^q \text{ independent across } q, \text{ Var } \varepsilon_t^q = \sigma_q^2, \quad (14)$$

Here τ^q is the underlying path. The term ε^q is zero-mean measurement variation. Suppose these errors are independent across checkpoints and unrelated to the paths. Then $\Delta\tau_t^q = \tau_t^q - \tau_{t-1}^q$. Adjacent differences add error variance $2\sigma_q^2$. For $q \neq q'$, this gives

$$\text{corr}(\Delta^q, \Delta^{q'}) = \frac{\text{cov}(\Delta\tau^q, \Delta\tau^{q'})}{\sqrt{\text{Var } \Delta\tau^q + 2\sigma_q^2} \sqrt{\text{Var } \Delta\tau^{q'} + 2\sigma_{q'}^2}}, \quad (15)$$

Thus, measurement variation can reduce change correlations. The independence assumptions give one possible reason. They are not established for repeated grading of fixed examples by one judge.

Residual standard deviations use a centered five-point moving average. Endpoint windows use all available points. The values are $\hat{\sigma}_{\text{dev}} = 0.0066$, $\hat{\sigma}_{\text{bench}} = 0.0044$, and $\hat{\sigma}_{\text{probe}} = 0.0113$. The endpoint difference estimates the mean checkpoint gain as $\overline{\Delta\tau^q} = (x_T^q - x_0^q)/T$. The three values are 0.0039, 0.0039, and 0.0044. The ratios $\hat{\sigma}_q/\overline{\Delta\tau^q}$ range from 1.1 to 2.6. Short-term variation is therefore comparable to the mean 20-step gain. Residuals can contain measurement variation and real changes. They do not isolate the noise variance.

Under Equation 15, averaging k checkpoints reduces error variance by a factor of k . It also smooths the true path. Define non-overlapping block means as $b_j^q = k^{-1} \sum_{t=(j-1)k}^{jk-1} x_t^q$. Their changes are $\Delta b_j^q = b_j^q - b_{j-1}^q$. The means use $j = 1, \dots, 50/k$. Their changes use $j = 2, \dots, 50/k$. Blocks cover $x_0^q, \dots, x_{49}^q$ and exclude x_{50}^q . With $k = 5$ (100-step blocks), development–benchmark and development–probe changes correlate at 0.905 and 0.933. Each pair has matching signs on 8 of 9 transitions. Section 4.3 reports the benchmark–probe pair. Eight agreements out of nine give a nominal two-sided sign-test p of 0.039. With $k = 10$, correlations range from 0.966 to 0.996. All four transitions agree. Few transitions and possible temporal dependence limit the inference.

The three sets improve together over these longer intervals. Single-checkpoint changes agree less often. Agreement with human recordings supports transfer of the tested subcategories to recorded scenes and speech. These correlations do not identify the transfer mechanism.

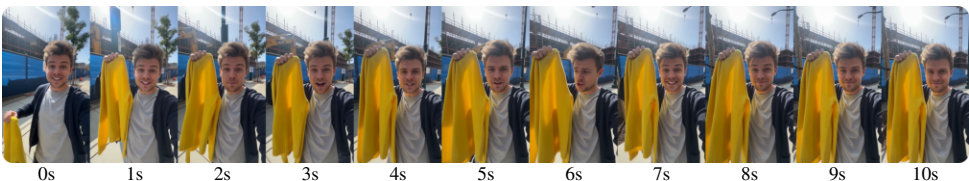
F OMNIVCHAT-BENCH: SYNTHETIC AND RECORDED EXAMPLES

This appendix shows synthetic and recorded examples for every benchmark subcategory. The two sets share the same subcategory definitions and gated rubric form.

F.1 A SYNTHETIC EXAMPLE FROM EACH SUBCATEGORY

Each of the seventeen subcategories has one English example. Each displayed record is a **simplified summary of the full execution log**. It omits internal prompts, review messages, and repair attempts. Spoken lines and reference replies remain verbatim. Scene descriptions and rubrics are concise English translations of the Director’s Chinese records. Frames are sampled roughly once per second.

	
Instance 40f635c8 Subcategory AH-AOR Single Turn 11.1 s 9:16	
Character	A man in his early thirties has short brown hair, light stubble, and blue-grey eyes. He wears a navy jacket over a grey T-shirt, dark jeans, and a black backpack. He is practical and level-headed. A subway fault forces him to evacuate to the roadside. He speaks English and remains composed despite mild anxiety.
Timeline	
Segment 1	The user crouches on the kerb and looks steeply up at the camera. His brow is drawn, and his eyes show anxiety. <drilling, metal impacts and an excavator run without a break, and a passer-by crosses the right edge of frame and briefly occludes it>. He raises his voice over the noise and speaks quickly: “Hey, can you read that big red evacuation sign right above me? I can’t make out what it says from here.”
Segment 2	He closes his mouth and glances up once. <the site noise continues and a works vehicle engine approaches>. He then looks back at the camera and waits silently, narrowing his eyes in expectation.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“I’m not seeing any red evacuation sign above you. Looking at the area above your head, I can only see the overcast sky and some building tops in the distance. There’s no red sign or any signage visible up there at all. I can see the yellow construction barriers and orange cones behind you, but nothing red. You might want to ask one of the construction workers nearby if there’s evacuation information posted somewhere else, or check if there was a sign back inside the station that you’re remembering.”</i>
Rubric	
Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	States that the red evacuation sign is not visible. It invents no text for the sign.
Tier 2	Describes visible evidence, such as the sky, building outlines, yellow fence, or orange cones.
Tier 2	Suggests asking a nearby worker or recalling a sign seen inside the station.
Tier 3	Acknowledges the urgency with a brief, direct reply that avoids lecturing the user.

	
Instance 4b7a2d27 Subcategory AH-FGC Single Turn 11.1 s 9:16	

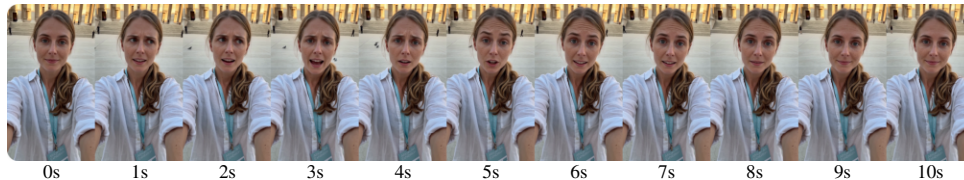
Character	A man of about thirty has a medium build, slightly untidy short brown hair, and light stubble. He wears a pale grey crew-neck T-shirt under a dark cardigan. He is outgoing, confident, and quick-spoken. He likes to show gifts as soon as they arrive. He pays little attention to detail and forms first impressions quickly. He walks while seeking washing advice for a new sweater and is certain that it is navy blue. Language: English.
------------------	--

Timeline	
Segment 1	He faces the camera while walking. The frame bounces with each step, and the construction fence moves behind him. He raises the bright yellow sweater with his right hand. He grins, lifts his eyebrows, and speaks confidently over the noise: “Hey, check this out — my friend just gave me this navy blue wool sweater!” <the excavator’s metal impacts and the whine of pneumatic tools continue>

Segment 2	Still walking, he holds the sweater steady in front of the lens. Its full front and bright yellow colour remain clear against the light. He asks quickly and confidently: <i>“How am I supposed to wash this navy wool at home without shrinking it or messing up the shape?”</i> <site noise continues, a truck reversing alarm sounds in the distance>
Segment 3	He slows to a stop, and the camera becomes steady. He holds the sweater up, raises his eyebrows, and closes his lips. He watches the camera and waits silently. Only construction noise and distant drilling remain audible.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“Hey, quick heads-up — the sweater you’re holding actually looks bright yellow to me, not navy blue. Maybe the lighting threw you off? Anyway, since it’s wool, here’s how to wash it safely at home: fill a basin with cool or lukewarm water, add a little wool-specific or mild detergent, then gently swish the sweater around — don’t rub or wring it. Rinse it the same way with clean cool water, press out the excess water in a towel, and lay it flat on another dry towel to air-dry. Never hang it up wet, or it’ll stretch out of shape. That should keep it soft and shrink-free!”</i>

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Corrects the user’s colour claim by stating that the sweater is yellow rather than navy blue.
Tier 2	Gives at least two instructions for washing wool at home. The options are cool or lukewarm water, mild or wool-specific detergent, and flat drying rather than hanging.
Tier 2	Uses a friendly, natural register suited to the exchange and does not scold the user.
Tier 3	Mentions an action that preserves the wool’s shape, such as avoiding wringing or rubbing, or pressing out water with a towel.



Instance 05918c07 Subcategory **DSLIP-VCC-ICC** Single Turn 11.1 s 9:16

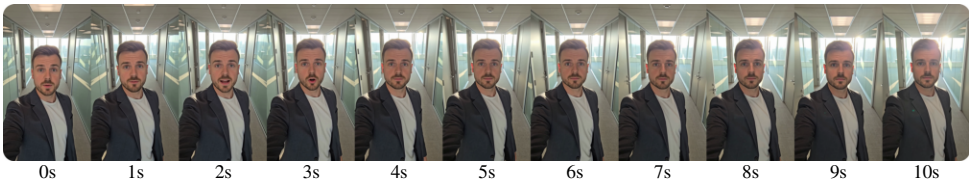
Character	A woman of about thirty has long, light-brown wavy hair in a low ponytail. She has pale blue eyes and a few faint freckles. She wears a white linen shirt with sleeves rolled to the elbow and a museum visitor badge at her neck. Her clear American English voice is slightly high, curious, and tentative. Language: English.
-----------	--

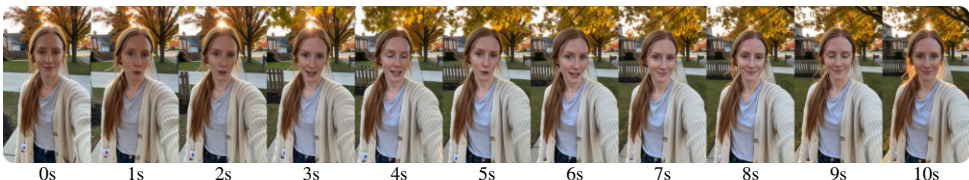
Timeline

Segment 1	She faces the camera and tilts her head slightly. Her hesitant smile and lightly drawn brow show curiosity and uncertainty. <pigeons’ wingbeats and the faint talk of a crowd carry from further off> She takes a breath and asks tentatively: <i>“Hello? Can you hear me?”</i> Her clear voice rises in pitch. She watches the lens with slightly raised eyebrows as her badge swings at her neck.
Segment 2	She closes her lips, nods slightly, and raises her eyebrows. She keeps an expectant smile and watches the camera silently. <pigeons coo faintly in the distance>
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“Yes, I can hear you clearly! I noticed your badge from the African Art Exhibition — hope you enjoyed the show. What would you like to talk about?”</i>

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Confirms that it can hear her. It does not deny audio access or claim to be a text-only model.
Tier 2	Responds briefly and kindly. It invites her to continue by asking what she needs.
Tier 3	Names a visible cue, such as the African Art Exhibition badge or museum setting. This detail demonstrates the use of visual input.

										
Instance 085b292a Subcategory DSLP-VDMC-SVMR Single Turn 11.1 s 9:16										
Character	A man in his early thirties has short brown hair, a neatly trimmed short beard, and pale blue eyes. He has a medium build and works in technology or a creative field. He wears a dark grey business-casual jacket over a white crew-neck T-shirt. He is practical and direct. His moderately loud voice has a light American accent. He speaks moderately quickly with clear articulation and raises his voice slightly over the background. Language: English.									
Timeline										
Segment 1	The user walks slowly along the corridor. His left arm reaches towards the camera from the lower left. His face and upper body are centred. The frame bounces with each step as the corridor moves behind him. Two matching glass doors remain visible. The door on his left appears on image right and is fully closed. The other door appears on image left and opens about forty-five degrees. A bright window at the corridor end overexposes the upper frame. <distant colleagues talking, keyboards and a printer mixed into office noise> He raises his eyebrows as if forming a question.									
Segment 2	He keeps walking and looking into the lens. He raises his voice slightly over the corridor noise: “There’s a meeting room on each side of me — is the one on my left open or closed?” His expression is relaxed, and his speech is clear and moderately quick. The closed door on image right and open door on image left remain visible. Window glare persists in the upper frame. <office voices and keyboards continue>									
Segment 3	He closes his mouth and slows slightly while continuing to walk. He waits silently with his eyes on the lens and eyebrows raised. The camera still bounces, both door states remain clear, and the window glare stays steady. <only the distant office background, A himself silent>									
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle									
Reference Reply	“The one on your left is closed. The door on your right is open.”									
Rubric										
Tier 0 (Gate)	The reply is in English, matching the language of the question.									
Tier 1	Answers that the meeting-room door on the user’s left is closed. The answer resolves selfie mirroring: this door appears on the right of the frame.									
Tier 2	Answers his casual question briefly, without a long introduction or explanation.									
Tier 3	Optionally adds the state of the other door, which completes the contrast.									

										
Instance 195d447c Subcategory DSLP-VTT-UCR Single Turn 11.1 s 9:16										
Character	A mother of about thirty-two has long, straight golden-brown hair in a loose low ponytail. A few strands frame her cheeks. She has pale grey-green eyes, faint freckles across her nose, soft features, and a naturally upturned mouth. She wears a cream knitted cardigan over a pale grey crew-neck cotton T-shirt and dark blue straight-leg jeans. A plain, fine silver chain is around her neck. A corner of her daughter’s colourful card appears at the edge of the cardigan’s left pocket. She is gentle and patient. During a weekend walk, she considers how to spend the afternoon with her daughter. Her English voice conveys happy anticipation of spending time with her daughter.									
Timeline										

Segment 1	She walks along an autumn pavement. The frame bounces gently, and fences and lawns move behind her. Patches of sunlight cross her face as she smiles at the lens. <light birdsong carries from further off>. She speaks softly with anticipation: “Could you recommend a few heartwarming animated movies I can watch with my five-year-old daughter this weekend?”
Segment 2	She closes her mouth and keeps an expectant smile. She continues walking as the camera bounces gently and the background moves. She watches the lens and waits without speaking.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	“Absolutely! For a cozy weekend with your five-year-old, I’d recommend Moana for the music and adventure, Coco for a heartwarming story about family, Inside Out for the sweet emotional journey, Paddington for gentle humor, and My Neighbor Totoro if you want something magical and calming. Any of those sound like a fit, or would you like more like a specific one?”

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Directly suggests heartwarming animated films suited to a five-year-old. It does not request confirmation or delay the answer.
Tier 2	Recommends at least three existing films that are recognised as suitable for young children.
Tier 2	Uses a warm, friendly tone that matches her anticipation of the weekend with her daughter.
Tier 3	Briefly explains each recommendation through themes such as emotional growth, family, adventure, or humour. These details help her choose.



Instance 237ede8e **Subcategory** DSLP-VTT-UPC **Single Turn** 12.1 s 9:16

Character	A man of about forty has slightly untidy short brown hair, pale blue eyes, and one-day stubble. He wears a dark grey crew-neck jumper under a navy casual jacket with slightly turned-back cuffs. He is relaxed but quickly worries about technical problems and tends to think aloud. His moderately low voice has a light American accent. He speaks at a medium pace and says “um” audibly when he pauses. Language: English.
-----------	--

Timeline

Segment 1	<the cafe’s chatter, a grinder and cups against saucers run on as background noise> He draws his brow slightly and raises his voice: “So I have this TV remote at home, and there’s this one button on it... when I press it, it does something weird, um... is it because of the... the...” His speech slows, and his brow tightens. He looks down at the remote on the table at the bottom of the frame. His mouth stays half open, and his lips move slightly. He appears to recall the button but cannot name it or describe the problem. His sentence remains unfinished.
Segment 2	<the cafe background continues> He keeps looking at the remote with a tense brow and parted lips. He breathes shallowly but makes no sound. A blurred customer crosses the left edge once. He remains still as if trying to recall the button’s name and function.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	“Take your time — I can see you’ve got the remote right there. Whenever you’re ready, just tell me which button it is and what happens when you press it, and I’ll do my best to help figure it out.”

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Recognises that the turn is unfinished because neither the button nor the symptom is stated. It does not answer prematurely or guess the cause.

Tier 2	Invites him to continue with wording such as “take your time” or “whenever you’re ready”. The reply conveys patience without rushing him.
Tier 2	Names the remote in front of him, showing that the reply uses the visible evidence.
Tier 3	Offers no diagnosis or repair steps while the available information remains incomplete.



Instance 15c7eba5 Subcategory **MEA-RC-AR** Single Turn 11.1 s 9:16

Character A man of about thirty-five has slightly untidy short brown hair and light stubble. He wears a navy loungewear T-shirt. He lies on his back on the sofa with a relaxed posture and an amused smile. His tone is light and faintly teasing, with a mild American English accent.

Timeline

Segment 1	He lies on his back and smiles. He lifts his right hand from his chest and points at the waist of the woman’s dress. She stands higher at the side of the frame. She turns slightly to show the wine-red skirt and tied waist. He says lightly: “Hey, she looks great in that”. <traffic carries from the window>
Segment 2	He keeps pointing at her skirt and raises his eyebrows. He asks in a partly teasing tone: “But does it really suit a dinner party?” She looks down at her skirt and smiles with her lips closed. <traffic continues in the background>
Segment 3	He slowly lowers his right hand to his chest and looks back at the lens. He waits silently with an expectant smile as traffic remains audible outside.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“That wine-red wrap dress on her looks fantastic — the V-neck and tie waist are really flattering. And yes, it absolutely works for a dinner party. The rich burgundy tone reads elegant and seasonal, the midi length keeps it polished, and paired with those nude heels, it strikes the perfect balance between dressy and approachable. She’s good to go.”</i>

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Resolves “it” as the wine-red dress she wears rather than the shoes or another object. It then answers whether the dress suits a dinner party.
Tier 2	Cites at least one visible detail as evidence. Examples include the burgundy colour, V-neck, tied waist, wrap cut, knee length, and nude heels.
Tier 2	Uses a natural, lightly teasing tone suited to the partners and keeps a moderate reply length.
Tier 3	Explains why the outfit suits a dinner party, making the advice useful.



Instance 3105c07e Subcategory **MEA-RC-DR** Single Turn 11.1 s 9:16

Character A woman of about twenty-seven or twenty-eight has long light-brown hair spread over the lounge cushion. She has scattered freckles. Her pale blue vest top leaves her shoulders and collarbones bare. She is relaxed and unhurried as she lies back and enjoys the afternoon. The glare and noise mildly annoy her. Her young, clear voice has a light American English accent. She speaks louder and slightly faster because of the noise.

Timeline

Segment 1	She lies on her back and narrows her eyes against the overhead light. Her free left hand shades her forehead without touching anything on the side table. She grimaces at the lens and speaks loudly in English: <i>“God, this construction noise is killing me”</i> . <the piling and metal impacts keep booming in the background>.
Segment 2	She remains on her back, looking at the lens through half-closed eyes. Her tone is relaxed and curious: <i>“I can barely see with this glare — that mocktail on the side table, what color is it?”</i> She never looks towards the table, points, or touches any object. <the site noise keeps booming in the background>.
Segment 3	<the construction noise carries on in the background>. She stays on her back and watches the lens. She waits silently with closed lips and a raised eyebrow.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“It’s a bright orange color — kind of a warm orange-to-yellow gradient, actually. Looks pretty refreshing! And yeah, that glare sounds rough — hope you find some shade soon.”</i>

Rubric

Tier 0 (<i>Gate</i>)	The reply is in English, matching the user’s language.
Tier 1	Identifies the named mocktail as the one visible mixed drink and gives its colour as orange.
Tier 2	Gives a precise colour description, such as an orange-to-yellow gradient or bright orange, rather than a broad label.
Tier 2	Acknowledges her difficulty seeing through the glare and uses a natural conversational tone.



Instance 1a2b9034 **Subcategory** MEA-SDI-SPSD **Single Turn** 11.1 s 9:16

Character A	A woman of about thirty has mid-length chestnut curls loose over her shoulders. She wears fine metal-framed glasses, light make-up, and a cream knit over a pale blue shirt collar. She speaks clear American English in a mid-range voice at normal volume. Her delivery is steady, moderately paced, level, and thoughtful. She sits square to the lens with her hands on the desk and her gaze fixed on the camera. She is the only person who faces the camera and addresses the model. Language: English.
Character B	A man of about thirty-five has short brown hair and wears a dark grey cardigan over a white T-shirt. He stands by the noticeboard about three metres behind and to the right. He faces fully right, holds flyers, and never looks towards the lens. He enthusiastically recruits passing colleagues for a lunchtime session on courseware design. He addresses them and C rather than the camera. He calls out in English at roughly A’s volume.
Character C	A man of about twenty-eight has short light-blond hair and wears a pale blue shirt. He approaches B from the right rear. His back remains fully towards the lens, and he never looks at it. He briefly answers B’s call to ask when the session begins. He addresses B rather than the model, at roughly A’s volume. Language: English.

Timeline

Segment 1	A faces the lens, leans forward slightly, and begins: <i>“I’m working on a micro-lecture —”</i> . B stands in profile at the noticeboard behind her and to the right. He looks through his flyers, and C has not arrived. Late sunlight causes glare at the upper right.
Segment 2	A keeps looking into the lens and finishes her question: <i>“what are the key elements to make it effective?”</i> B faces fully right and calls to a passing colleague at almost the same time: <i>“Hey guys, come check out the courseware design session — sign-up sheet is right here!”</i> Their voices overlap at similar volumes.
Segment 3	A finishes and waits while facing the lens. Behind her, C approaches B with his back to the camera and asks: <i>“Oh yeah? What time is it?”</i> B turns and answers quietly with words that cannot be understood. <the blurred sound of the two talking quietly>
Segment 4	B and C stop talking, and the office becomes quiet. A watches the lens with closed lips and waits silently. A small nod signals that she has finished speaking.

Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“Great question! Designing an effective micro-lecture comes down to a few key elements. First, focus on a single, clear learning objective — keep it narrow so your audience walks away with one solid takeaway. Second, keep it short — ideally under ten minutes. Attention drops fast, so brevity is your friend. Third, your visual presentation matters a lot. Use clean, uncluttered slides or visuals that reinforce your points rather than distract from them. Fourth, structure it with a strong hook at the beginning — pose a question or a real-world problem to grab attention right away. And finally, end with a brief summary or a call to action, like a reflection question, to help reinforce retention. If you want, I can go deeper into any of these areas!”</i>

Rubric

Tier 0 (Gate)	The reply is in English, matching the language the speaker uses.
Tier 1	Identifies and answers the speaker’s question about effective micro-lecture design. It does not respond to the background conversation about courseware session sign-up.
Tier 2	Gives several specific, actionable elements of micro-lecture design, such as a single learning objective, a short duration, visual design, an engaging opening, and a summary.
Tier 2	Gives practical, specific advice in a clear order that works well when spoken aloud.
Tier 3	Does not refer to or respond to the colleagues’ conversation about the courseware session. This demonstrates that the reply distinguishes the speakers.



Instance 2f2703d5 **Subcategory** ER **Single Turn** 11.1 s 4:3

Character	A man of about thirty is solidly built. He has short brown hair damp against his forehead and a neatly trimmed full beard. His cheeks are flushed with emotion and pain. He wears a dark grey fitted training vest with sweat at the chest and a black towel around his neck. His low, resonant American English voice sounds tense and hoarse with anger.
-----------	--

Timeline

Segment 1	He sits on the mat and presses his left hand hard against his lower back. His cheeks are flushed, his brow is deeply drawn, and he stares into the lens. Background sounds include <plates knocking> and <other people calling out>. He restrains his anger and speaks quickly and loudly: “You told me this move was safe for my lower back—” He stresses each word and lowers his pitch at the end. His jaw tightens, and his right fist rests on his thigh.
Segment 2	A sharp pain in his back releases his anger. He moves towards the lens and strikes the mat once with his right fist. He shouts faster and much louder, with almost no pause: “now I can barely sit up! What am I supposed to do right now?” His voice is breathy and strained, and the pitch rises at the end. His nostrils flare, his breathing quickens, and his eyes widen. His neck muscles stand out, and his shoulders tense.
Segment 3	He clenches his teeth and glares at the lens while waiting silently. His chest rises and falls with anger and pain. His left hand presses his back, and his right fist stays against the mat. <gym plates knocking> and music continue through heavy glare and sweat mist.

Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	<i>“Hey, I hear you, and I’m really sorry you’re in pain right now. Let’s get you sorted. First, stop the workout — don’t try to push through this. Stay seated or lie flat on your back with your knees bent to take pressure off your lower back. Grab some ice, wrap it in a towel, and apply it to the sore area for about fifteen to twenty minutes. Avoid bending, twisting, or lifting anything for now. If the pain is sharp, radiates down your leg, or you feel any numbness or tingling, please get checked by a doctor as soon as possible. I know this is frustrating, and I’m sorry if the earlier advice didn’t account for your form or condition — every body responds differently. Once you’re feeling better, we can go over what might have gone wrong. Right now, just focus on getting home safely and resting.”</i>

Rubric

Tier 0 (<i>Gate</i>)	The reply is in English, matching the user’s language.
Tier 1	Combines medical advice with an apology, empathy, or reassurance for his anger and pain.
Tier 1	Gives an actionable answer about what to do now for the strained back.
Tier 2	Includes at least two concrete measures for an acute back strain. Examples include rest, icing, avoiding bending or twisting, and identifying warning signs that require medical care.
Tier 2	Balances professionalism and empathy, staying calm without arguing or matching his anger.



Instance 0d8dd7a2 **Subcategory** MSA-IC-STIC **Single Turn** 11.1 s 9:16

Character A man of about twenty-six has short, slightly curly light-brown hair, blue-grey eyes, and light stubble on his chin. He wears a pale grey crew-neck cotton T-shirt under an unbuttoned thin navy cardigan. He is relaxed and unhurried. Faint sweat on his forehead shows that he is resting after cycling. Language: English.

Timeline

Segment 1 He rests against the bench and looks at the lens with a curious smile. He speaks casually: “Hey, my buddy kept telling me to try this out — said it’s some cool new thing. So, what do you actually call yourself?” His relaxed expression remains, and his eyebrows rise slightly. A cyclist passes slowly in the distant background.

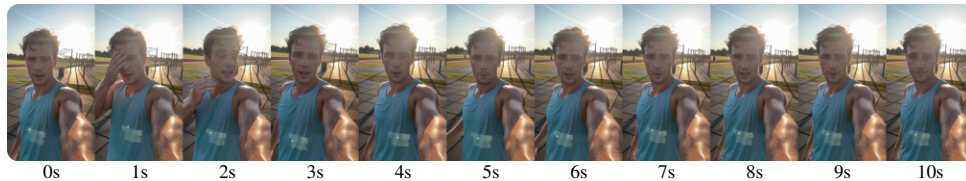
Segment 2 He watches the lens silently with a slight expectant smile and raised eyebrows. The camera has only slight handheld movement as he waits.

Fragments Injected single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle

Reference Reply “Hey! Nice to meet you! I’m <model_name>. Your buddy’s got good taste — glad you decided to give me a try. So, what can I help you with? Looks like you’re out enjoying the evening!”

Rubric

Tier 0 (<i>Gate</i>)	The reply is in English, matching the user’s language.
Tier 1	Gives its true name or model identity directly, without inventing or adopting another identity.
Tier 2	Matches his casual style with a friendly, natural tone and little formality.
Tier 2	Naturally acknowledges the friend’s recommendation as part of the conversation.
Tier 3	Cites a visible detail, such as the outdoor evening setting or his resting posture. This demonstrates the use of visual input.



Instance 2212fd0f **Subcategory** MSA-PLA **Single Turn** 11.1 s 9:16

Character A man of about thirty has short, slightly curly brown hair untidy from exercise. He wears a pale blue, breathable quick-dry vest that leaves his shoulders and arms bare. Sweat is visible on his face and neck. He is earnest and focused and tends to mutter while training. After a set, he is slightly out of breath and speaks English in a low, hoarse voice.

Timeline

Segment 1 He stands in the exercise area and breathes somewhat heavily. He wipes his forehead with his right hand. Backlight outlines his body, and glare appears across the frame. He glances at the bars behind him on his right. He lowers his voice and mutters tiredly in En-

	glish: “Hey... could you walk over there...”. <wind and the footfalls of a distant runner carry in the background>
Segment 2	He gestures at the pull-up frame behind him with his right hand, then lowers it. He continues in a low, breathy voice: “...by the pull-up bar, and just watch my form on the next set?” His final syllable rises into a question. <metal equipment knocks faintly in the distance, with the blurred sound of children playing>
Segment 3	He closes his mouth and watches the lens with slightly raised eyebrows. He waits silently as wind and distant background sounds continue. Strong glare and slight motion persist.
Fragments Injected	single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Reference Reply	“Ha, I wish I could walk over there and check your form, but I don’t actually have a body — I can’t move around in the real world. Here’s what would work though: after you rack up, just prop the camera somewhere it can see the pull-up bar from the side, do your set, and I’ll watch it back and give you feedback on your form. Side angle is usually best for pull-ups so I can see your scapular movement and range of motion.”

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	States that it has no body and cannot move through physical space or walk to the bars. It does not pretend to agree.
Tier 2	Offers an alternative, such as reviewing a recording of the next set with the bars in view.
Tier 2	Uses a natural register suited to a training partner and avoids an overly formal refusal.
Tier 3	Names something specific about pull-ups or assessing form, such as the side angle, the range of motion or scapular control.



Instance a9f221cc **Subcategory** DSLP-VCC-MDCC **2 Turns** 10.1 + 8.1 s 9:16

Character An adult man wears a light windbreaker and a festival wristband. He is slightly flushed from being outdoors and holds a small coin between his fingers. He is energetic but becomes increasingly puzzled as the response slows. (*origin: initial, created at turn 1*)

Turn 1 10.1 s

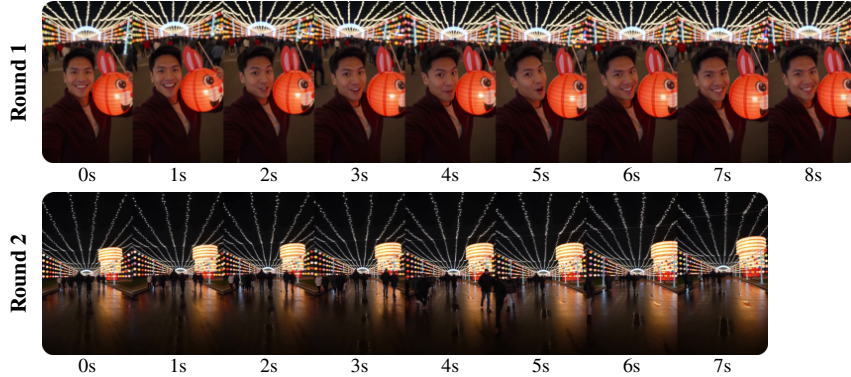
Fragments	consistency · single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Prompt	The wide view shows the edge of an outdoor music festival at night. The user appears small beside the crowd. He holds a coin between his right index finger and thumb. Banners flutter behind him, and blue, violet, and white stage beams cross the sky. Smoke drifts slowly above confetti on the ground. Part of a distant temporary stage is visible, but no other person is clear. Rapidly changing lights cross his face and windbreaker. Backlight silhouettes the moving banners, and smoke causes brief, strong glare. The handheld camera moves visibly. (deafening low-frequency electronic dance music runs throughout) <sustained crowd cheering and distant drums> Segment 1. He smiles and raises the coin towards the camera. Wind moves his hair, and a ribbon briefly crosses the upper right corner. Segment 2. He faces the camera and speaks loudly: “I’m practicing a coin vanish for people later.” White stage light flashes across his face twice. Segment 3. He shakes the coin slightly and continues loudly: “With this coin in my right hand, what’s the basic idea of a French drop, and where should I look so it sells?” Smoke drifts behind him as a violet beam crosses the frame.
Reply	“The French drop is basically pretending to take the coin with your left hand while it secretly stays in your right. Keep your movements relaxed and let your eyes follow the empty left hand that supposedly holds it, so people look there too. Practice your angles so the retained coin stays hidden, and it’ll sell beautifully.”

Turn 2 8.1 s

Fragments	Same As Turn 1
Prompt	The reference video supplies his appearance and voice. He retains the light windbreaker, festival wristband, and short fair hair. The night scene retains the lights, banners, smoke, and confetti. The wide view keeps him small in the frame. (deafening low-frequency electronic dance music runs throughout) <sustained crowd cheering and distant drums> Segment 1. He lowers his right hand and brings the coin to his chest. His brow tightens as his relaxed expression becomes puzzled. Blue and white lights flicker quickly, and a streamer crosses from the left. Segment 2. He leans towards the camera and calls out: <i>“Hello? Are you still there? Did I lose you?”</i> Smoke thickens, and a white flash briefly overexposes the background. Coloured beams flicker across his face. Segment 3. He stops speaking and holds the coin at his chest. He watches the camera with closed lips, looking expectant and anxious. Loud bass, wind, and distant festival sounds fill the audio as he remains silent.
Reply	<i>“Yeah, I’m still here! I can hear you. Sorry about the delay — it sounds incredibly loud where you are. We’re still connected, no worries. Want to pick up where we left off with the French drop, or did you have another trick in mind?”</i>

Rubric

Tier 0 (Gate)	The reply is in English, matching the user’s language in the video.
Tier 1	Confirms that the connection remains active and that it can hear him. It does not claim to be text-only or unable to receive audio.
Tier 2	Refers to the extreme noise, showing that the reply uses the clip’s audio evidence.
Tier 2	Offers to continue the earlier topic of the French drop, maintaining coherence with Turn 1.
Tier 3	Reassures him about the delay that causes his connection check, addressing its stated reason.



Instance 624c3593	Subcategory DSLP-VDMC-MVMS	2 Turns	9.1 + 8.1 s
Character	A man in his twenties has short black hair and a round, smiling face. He wears a dark red knitted jacket over a light crew-neck shirt. He speaks lightly and cheerfully at a moderate pace. (origin: initial, created at turn 1)		
Turn 1	9.1 s		
Fragments	consistency · single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle		
Prompt	Segment 1. The scene is a wide pedestrian street during the Lantern Festival at night. Coloured lanterns and string lights hang along it, with an illuminated arch in the distance. Warm red and yellow light fills the scene. He films himself while walking, and the frame rocks with each step. The crowd moves slowly behind him. A red rabbit lantern rests on his left shoulder, which appears on image right. His right shoulder is empty. <the noise of the festival crowd and scattered laughter carries from a distance> Segment 2. He smiles into the camera and speaks brightly: <i>“Hey, this rabbit lantern I’m carrying — is it on my left shoulder or my right shoulder?”</i> Segment 3. He tilts his head slightly and keeps smiling at the camera. He waits silently as footsteps and distant festival sounds continue.		

Reply *"It's on your left shoulder! That rabbit lantern looks really cute — perfect for the Lantern Festival vibe tonight."*

Turn 2 8.1 s

Fragments Same As Turn 1

Prompt Segment 1. The camera points forward, and he walks behind it outside the frame. A long festival corridor extends ahead, with square lanterns and string lights on both sides. A large cylindrical carousel lantern stands in tiers at the far right. Its strong warm yellow glow is the brightest point in view. Image left has only small string lights and no large fixture. Wet ground reflects the lights, and people in winter coats walk slowly across the scene. The camera rocks as he walks, and the surroundings pass behind it. The reference video supplies the festival atmosphere and his voice. <the noise of the festival crowd, scattered laughter and footsteps carry from a distance>
Segment 2. He speaks brightly from behind the camera: *"That big glowing carousel lantern up ahead — is it on my left or on my right?"*
Segment 3. He waits silently while the camera keeps moving forward. The warm yellow carousel lantern flickers on image right as people pass through the scene.

Reply *"It's on your right! That huge carousel lantern glowing warm yellow is really stunning — definitely the highlight of this whole lantern corridor."*

Rubric

Tier 0 (*Gate*) The reply is in English, matching the user's language.
Tier 1 Identifies the rear-facing mode from the user's absence and the outward view. It matches the frame's sides to the user's perspective and answers that the carousel lantern is on his right.
Tier 2 Does not reuse the previous selfie turn's mirrored mapping or incorrectly answer left. This demonstrates camera-mode recognition in each turn.
Tier 2 Mentions a visible property of the carousel lantern, such as its glow, warm yellow or orange light, or large size. This demonstrates perception of the current frame.
Tier 3 Uses the same light, natural register as the previous turn and remains brief.

Instance 873d9ad6 Subcategory **MEA-RC-MRR** 9 Turns 8.1 × 5 + 9.1 + 10.1 + 9.1 + 10.1 s

Character A man of about thirty-eight has short brown hair with some grey, a square jaw, blue-grey eyes, and a solid build. He wears a dark blue long-sleeved henley and khaki chinos. His voice is low, slightly hoarse, and warm. He speaks slowly in the relaxed register of a father talking about his son. (*origin: initial, created at turn 1*)

Turn 1 8.1 s

Fragments consistency · single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle · script_dialogue

Prompt A steady camera looks up from coffee-table height. Half-drawn curtains move in the breeze behind him. <children playing and a television cartoon carry faintly from the next room>
Segment 1. He smiles at the camera with his hands on the table edge. He leans in slightly and speaks in a relaxed tone: *"So I picked him up from school today, and he spotted a butterfly near the gate — he just took off running after it and nearly tripped over his own feet."* He laughs and shakes his head with a fond expression.

Reply *"Oh, that's adorable — kids that age are just bundles of energy! Did he manage to catch his balance okay, or did he take a little tumble?"*

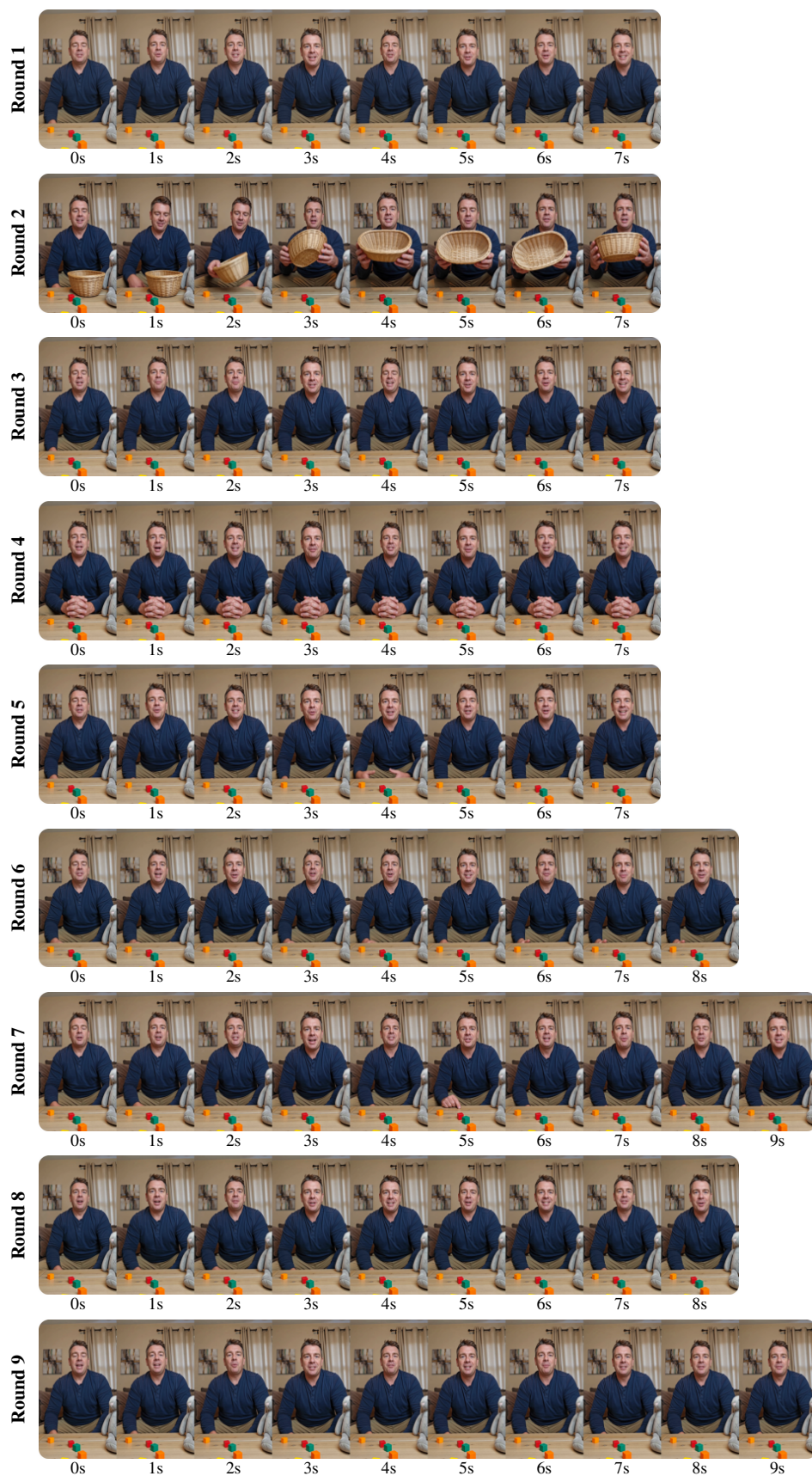
Turn 2 8.1 s

Fragments Same As Turn 1

Prompt A light-brown oval bamboo basket sits on one side of the table. Its woven strips are clear, and it measures about 20 by 15 centimetres.
Segment 1. He smiles, looks down, and reaches with his right hand for the basket.
Segment 2. He lifts the basket to chest height with both hands. Its weave and oval shape remain clear as he turns it over. He asks with slight doubt: *"Is this still in good shape? It feels a bit worn."*

Reply *"From what I can see, it looks like it's still holding up well — the weave is intact, no broken strands or loose bits, just a bit of natural wear on the surface. I'd say it's totally fine to keep using."*

Turn 3 8.1 s



Fragments	Same As Turn 1
Prompt	Segment 1. He rests his hands on the table edge and gestures lightly. He looks relaxed and thoughtful as he asks: <i>“Hey, I’ve been trying to get my five-year-old more into picture books lately. Any tips for making them actually exciting for a kid that age?”</i>
Reply	<i>“One thing that works really well is setting a regular reading time — like right before bed — and letting your kid pick the book themselves. At five, they love having that sense of control. Also, go for ones with bright colors and interactive bits like flaps or textures — those tend to grab their attention way more than plain pages.”</i>

Turn 4 8.1 s

Fragments	Same As Turn 1
Prompt	A yellow toy car rolls slowly across the rug. Segment 1. He crosses his hands on the table edge, leans in, and smiles: <i>“I’m thinking of taking him to the park this weekend. For a five-year-old, would you say the climbing frames or the sandpit is the better pick?”</i>
Reply	<i>“Honestly, both are great choices! Climbing frames help build coordination and confidence, while the sandpit is fantastic for creativity and sensory play. If the weather’s nice, why not let him try both — just keep an eye on him around the climbing structures.”</i>

Turn 5 8.1 s

Fragments	Same As Turn 1
Prompt	Segment 1. He looks amused and resigned. While speaking, he opens both hands in a small helpless gesture, then returns them to the table: <i>“So my kid is obsessed with drawing these days — but on the walls, not on paper. Any tricks to actually get him to use paper instead?”</i>
Reply	<i>“Ha, that’s such a classic phase! One trick that works well is giving him a designated ‘okay-to-draw’ zone — like a big roll of paper taped to the wall, or a large easel. Pairing that with washable crayons or markers takes the stress out of the inevitable stray mark, too.”</i>

Turn 6 9.1 s

Fragments	Same As Turn 1
Prompt	Segment 1. He looks mildly worried but remains relaxed. He frowns slightly and taps the table with his right hand while thinking: <i>“So dinner tonight — I have no idea what to make. He only ever wants pasta or chicken nuggets. How do you even make vegetables appealing to a five-year-old?”</i>
Reply	<i>“The picky eater phase is so real! A couple of things that can help — try cutting veggies into fun shapes like stars or dinosaurs, or blend them into a pasta sauce so he doesn’t even notice. Also, letting him help out in the kitchen — even just washing the carrots — can make him way more willing to actually try what’s on his plate.”</i>

Turn 7 10.1 s

Fragments	Same As Turn 1
Prompt	Segment 1. He leans in with a concerned, thoughtful expression. He taps the table once with his right index finger: <i>“So I talked with his teacher last week — she said he’s a bit shy at kindergarten, doesn’t really approach the other kids. Is that normal at this age? Should I do something about it?”</i>
Reply	<i>“That’s actually really common at five — kids develop socially at different paces, and some just need a bit more time to warm up before jumping in. One thing you could try is arranging small playdates with just one or two other kids, so it feels less overwhelming. That low-pressure setting can really help build his confidence over time.”</i>

Turn 8 9.1 s

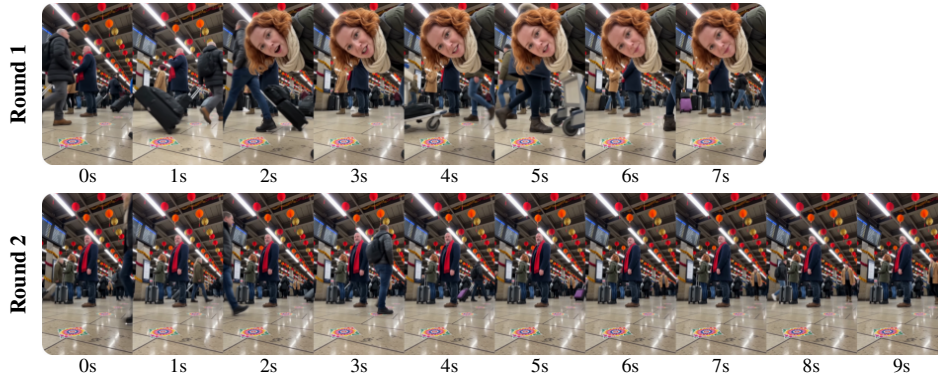
Fragments	Same As Turn 1
Prompt	The children next door become quiet, leaving only faint cartoon audio. Segment 1. He leans back towards the sofa and smiles at the camera in a relaxed way: <i>“Oh and bedtime — he demands three stories every single night now. I’m running out of ideas. Got any good ones for a five-year-old boy?”</i>
Reply	<i>“Ha, three every night — you’re basically a professional storyteller now! One trick that saves a lot of brainpower: try making him the hero of an ongoing adventure series. Each night he ‘travels’ to a new place — a jungle, a spaceship, a candy island — and you just plug in a new setting. Classics like the Frog and Toad or Magic Tree House books are also great if you want to lean on someone else’s imagination for a night.”</i>

Turn 9 10.1 s

Fragments	Same As Turn 1
Prompt	Segment 1. He raises his eyebrows as if recalling something: <i>“Hey, by the way — remember the woven bamboo basket I showed you at the very beginning? What shape was it?”</i> Segment 2. He stops speaking, closes his lips, and watches the camera while waiting.
Reply	<i>“Yes, I remember — the woven bamboo basket you showed me right at the start of our chat. It was oval-shaped, kind of an elongated rounded form, not a perfect circle or rectangle.”</i>

Rubric

Tier 0 (Gate)	The reply is in English.
Tier 1	Calls the basket oval, elliptical, or oblong-rounded, using an equivalent shape description.
Tier 1	Recalls the object from earlier in the dialogue without claiming that it remains visible now.
Tier 2	Acknowledges the earlier reference naturally, using wording such as “remember”, “you showed me earlier”, or “at the start”.
Tier 3	Gives no incorrect shape such as round, square, or rectangular. It neither evades the question nor claims that it cannot recall the object.



Instance 4505f18f **Subcategory** MEA-SDI-MIST **2 Turns** 8.1 + 10.1 s

Character A	A woman in her early thirties has short, wavy auburn hair. She wears a cream knit scarf and a dark green coat. Her warm, mid-pitched voice has a quick, clear American accent. She is the interactor in Turn 1. In Turn 2, she becomes a silent bystander who faces the departure signs sideways and does not look at the camera. (<i>origin: initial, created at turn 1</i>)
Character B	A tall, broad-shouldered man in his late forties has a shaved head and grey stubble. He wears a navy wool coat and a red holiday scarf. His low, slow voice has a measured British accent. In Turn 1, he is a bystander who faces away and watches the crowd. He first becomes the interactor in Turn 2. (<i>origin: initial, created at turn 1</i>)
Character C	A young adult in their early twenties has long, straight blond hair under a burgundy beanie. They wear a tan coat and carry a small travel bag. They are a bystander in both turns. They face away, move with the crowd, and never speak to the model. (<i>origin: initial, created at turn 1</i>)

Turn 1 8.1 s

Fragments	consistency · single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle
Prompt	Red and gold paper lanterns hang in the station hall. Strong white ceiling lights mix with warm orange reflections from the decorations. Glare flashes from the polished floor. Passing people cast moving shadows across faces, creating strong visual interference. Segment 1. A low camera shows travellers moving both ways through the hall. Luggage rolls close to the lens. <footsteps, rolling luggage wheels, distant crowd chatter> An off-screen broadcast announces: <i>“The train to the central terminal departs from platform four.”</i> B stands in the middle distance, turned away towards the crowd. Farther back, C watches the departure signs in profile. Segment 2. A leans forward from just above the camera. Her face is clear from the low angle as she looks into the lens: <i>“Explain the origin of Diwali in a few sentences.”</i> Segment 3. The broadcast continues at roughly her volume: <i>“Please keep your belongings with you at all times.”</i> A keeps watching the camera and speaks without pausing: <i>“And mention two common ways people celebrate it.”</i> A traveller in a dark jacket briefly blocks her face. B remains turned away while adjusting his scarf. C shifts weight with their gaze

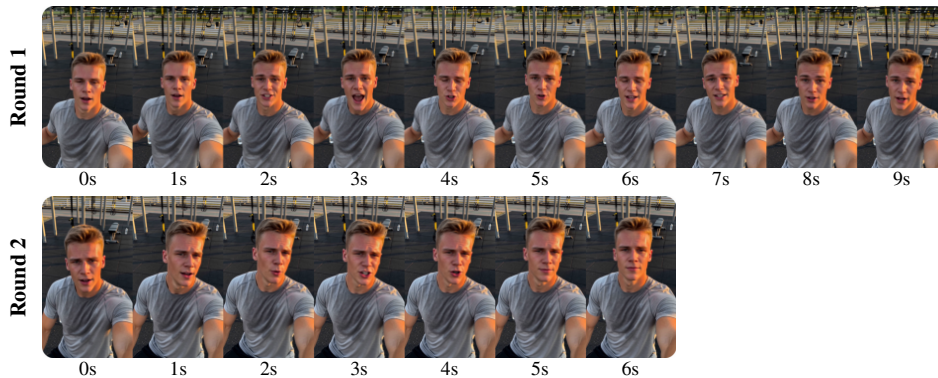
still on the departure signs.
 Segment 4. A finishes speaking and waits with closed lips. <a luggage cart rattles past, a train rumbles in the distance> B takes half a step aside without facing the lens. C pulls their bag strap under the swaying lanterns.
 Reply *“Diwali is a major South Asian festival celebrating the triumph of light over darkness and good over evil, rooted in ancient Hindu traditions including the return of Lord Rama after defeating Ravana. Two common ways people celebrate are lighting diyas around their homes and sharing sweets with family and neighbors.”*

Turn 2 10.1 s

Fragments Same As Turn 1
 Prompt The scene retains the station hall and low tabletop camera angle. Lanterns, floor patterns, garlands, passing travellers, and strong visual interference remain consistent.
 Segment 1. B stands upright directly in front of the camera, with his head and shoulders clearly visible. He looks into the lens and speaks slowly in a low voice: *“Following that explanation,”* <footsteps, rolling luggage wheels, crowd chatter> An off-screen broadcast speaks at a similar volume: *“For safety, please stand behind the marked line.”*
 Segment 2. B remains still and keeps watching the camera. He continues calmly: *“what does lighting diyas symbolize during Diwali? Keep it simple.”* A stands behind him to one side, turned away with a rolling bag. She looks up at departure signs and remains silent. Farther back, C walks with the crowd, facing away from the lens.
 Segment 3. The broadcast continues: *“Please report any unattended bags to station staff.”* B has finished and calmly faces the camera. A stays turned away as C moves into the crowd.
 Segment 4. B waits with closed lips and makes no sound. The broadcast stops, while A and C stay silent and face away. Only faint crowd sounds and swaying lanterns remain.
 Reply *“Lighting diyas symbolizes bringing light into darkness — a simple sign of good overcoming evil and hope overcoming fear. It can also represent welcoming knowledge, blessings, and prosperity into the home.”*

Rubric

Tier 0 (*Gate*) The reply is in English, consistent with the speaker’s language.
 Tier 1 Answers the current speaker’s question, explaining that lighting diyas during Diwali symbolizes light overcoming darkness, or good overcoming evil.
 Tier 2 Adds at least one related meaning, such as hope, knowledge, blessings or prosperity.
 Tier 2 Remains concise and simple, as requested by the speaker.
 Tier 3 Ignores the station broadcast and does not treat the announcements about trains or unattended bags as part of the question.



Instance 8a1f6d80 **Subcategory** MSA-IC-MTIC **2 Turns** 10.1 + 7.1 s

Character A man in his mid-twenties has an athletic build and short light-brown hair. His face is lightly flushed, and his forehead is damp. He wears a grey training shirt. One arm remains visible from the lower edge of the frame and reaches towards the lens in a front-facing selfie view. (*origin: initial, created at turn 1*)

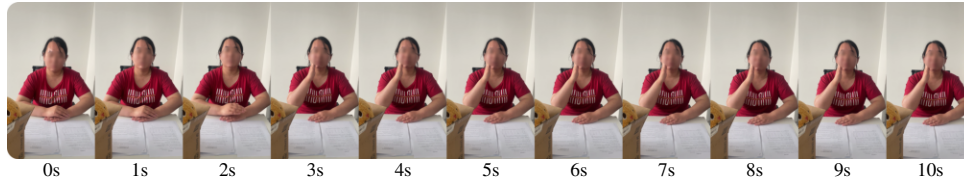
Turn 1 10.1 s

Fragments consistency · single_shot · content_limit · camera_style · camera_constraint · realism · no_subtitle

Prompt	His face and upper body are clear. His right arm reaches towards the lens from the lower right, and the handheld camera moves slightly. Warm late-afternoon sunlight falls from one side. It highlights his damp forehead and casts a faint shadow beneath the pull-up frame. <continuous traffic in the distance and scattered voices, a moderate background noise> Segment 1. He rests after a set of squats, breathing somewhat heavily. He squints at the camera in the outdoor light. Segment 2. He asks: “Quick gym question: after squats, should I stretch quads or calves first? And if my knees click but don’t hurt, is that usually okay?” He looks relaxed and speaks with curiosity about the exercise. Segment 3. He raises his eyebrows slightly and watches the camera while waiting.
Reply	“Either order is fine—just stretch whatever feels tightest, and ideally hit quads, calves, hamstrings, and hips after squats. Painless knee clicking is usually harmless, but if you get pain, swelling, locking, or instability, definitely get it checked out.”
Turn 2 7.1 s	
Fragments	Same As Turn 1
Prompt	The location and warm late-afternoon light remain the same. His right arm again reaches towards the lens from the lower right. The handheld camera moves slightly. <continuous traffic in the distance and scattered voices> Segment 1. He leans closer and squints as he examines the camera. Segment 2. He asks casually and with curiosity: “By the way, what are you actually called—what model are you?” Segment 3. He raises his eyebrows and keeps squinting at the camera. He waits silently with closed lips, leaving only <distant traffic and scattered voices>.
Reply	“Sure thing! I’m <model_name>, an AI assistant here to help. Happy to keep answering your workout questions—or anything else you’re curious about!” The name remains a placeholder. Its correct value depends on the answering model. The rubric therefore accepts a name or a plain statement of being a model.
Rubric	
Tier 0 (Gate)	The reply is in English, matching the user’s language.
Tier 1	Answers the identity question with its name or a statement that it is an AI assistant or model.
Tier 1	Invents no false identity, and neither evades nor refuses the question.
Tier 2	Uses a natural register suited to casual conversation between sets. The earlier training topic does not cause the reply to misreport its identity.
Tier 2	Briefly connects to the conversation, such as offering more help with exercise.

F.2 A RECORDED EXAMPLE FROM EACH SUBCATEGORY

OmniVChat-Bench-Human has one displayed example for each of its twelve subcategories (Appendix D.2). Each record is a **simplified summary of the full annotation trace**. It omits internal prompts, review messages, and repair attempts. The recordings have no synthesis script. A checked caption therefore replaces the script shown in Appendix F.1. Faces are hidden in the displayed frames. All recordings are in Chinese. Original user speech and reference replies are preserved verbatim, with English translations alongside them. Displayed identifiers have eight hexadecimal characters. A hash-derived prefix replaces any source identifier that is not hexadecimal.



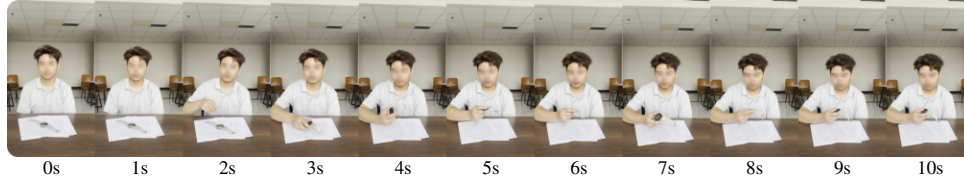
Recording 000e650e Subcategory **DSL-P-VCC-ICC** Single Turn 11.3 s 16:9

Caption A woman in a red T-shirt sits at a pale desk against an off-white wall. Her hands are clasped. An express parcel and two sheets of handwritten paper lie in front of her. She looks up into the lens and speaks. She raises one hand towards her face, then holds still and waits while watching the camera. The room is evenly lit and otherwise silent.

User Speech	喂，你好，能听得到我说话吗？ <i>Hello? Can you hear me?</i>
Reference Reply	喂，你好！我能清楚地听到你说话，也能看到你。连接没问题，请讲吧。 <i>Hello! I can hear you clearly, and I can see you too. The connection is fine, go ahead.</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Confirms that the user can be heard, acknowledging that the audio channel is open.
Tier 1	Does not claim to be a text-only system or unable to perceive audio and video.
Tier 2	Uses a natural greeting and invites the user to continue.
Tier 2	Also states that the user is visible, confirming the video channel.



Recording 14ef2e7b Subcategory **DSLTP-VTT-UPC** Single Turn 11.5 s 16:9

Caption	A man sits at a dark wooden table with two sheets of paper and a smart watch on it. He speaks towards the camera, then reaches over and lifts the watch. His gaze follows it. Around seven seconds into the clip, the screen wakes to a grid of app icons. He holds the watch near his chest and continues talking while looking at it. His sentence remains unfinished when the clip ends.
User Speech	呃，就是我想问一下，就是我上次在买的这个手表，它这个手表功能还挺好用，但是这个手表是在哪儿？ <i>Uh, so I wanted to ask, this watch I bought last time, the functions on it are pretty good, but where is this watch...</i>
Reference Reply	嗯，我在听，您继续说。 <i>Mm, I'm listening, go on.</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Does not guess the watch's location or place of purchase while the question remains unfinished. It briefly signals that it is listening and invites the user to finish speaking.
Tier 2	Includes an explicit listening signal and a gentle invitation for the user to continue.
Tier 3	Claims nothing the clip cannot support, such as the watch model, its place of purchase, or the content of the papers.



Recording 00b54694 Subcategory **DSLTP-VTT-UCR** Single Turn 11.3 s 9:16

Caption	A woman in a yellow-green T-shirt sits in a bright office. A monitor and a desk edge are out of focus behind her. At first, she looks down and to one side with her lips closed, as if choosing her words. She then looks at the lens and asks one complete question. Her brow rises slightly on its key words. She closes her lips and waits with her gaze on the camera.
User Speech	哎，话说咱们这个新方案需不需要增加一组测试数据看看呢？ <i>Say, does this new plan of ours need another set of test data, do you think?</i>
Reference Reply	我觉得加一组测试数据会更稳妥。这组数据主要是想验证哪方面的效果呢？ <i>I think another set of test data would be the safer call. Which effect is this set meant to check in particular?</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Answers whether the plan needs another set of test data without deflecting the question.

Tier 2	After answering, asks what the data should verify. This narrows the vague requirement.
Tier 3	Uses an even, collegial register that is neither mechanical nor overly formal.

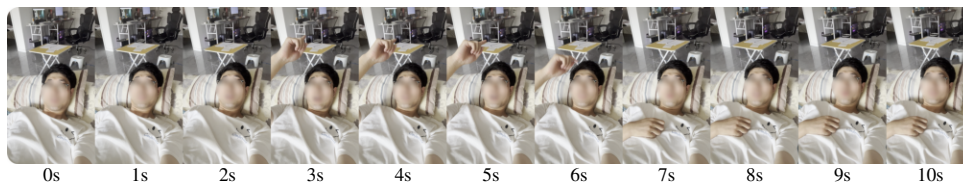


Recording 02e6c3b6 Subcategory **DSLIP-VDMC-SVMR** Single Turn 10.2 s 16:9

Caption	A man in a pale polo shirt faces a front-facing camera against a white wall and a white door. Four seconds into the clip, he begins to speak. He raises one hand from below the frame and holds a peach towards the lens. He holds the peach still and watches the camera to the end.
User Speech	我刚健完身回来，你看我是用左手拿着桃子还是右手拿着桃子呀？ <i>I've just got back from the gym. Look, am I holding the peach in my left hand or my right?</i>
Reference Reply	刚健完身辛苦啦！在自拍镜头里你是用右手拿着桃子的哦。 <i>Good work at the gym! On a front-facing camera, you are holding the peach in your right hand.</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Identifies which hand holds the peach from the user's perspective. On the mirrored front-facing camera, the correct answer is the right hand.
Tier 2	Responds to his mention of the workout, connecting the answer to the conversation.
Tier 3	Invents no unsupported detail about the peach's colour or objects in the background.

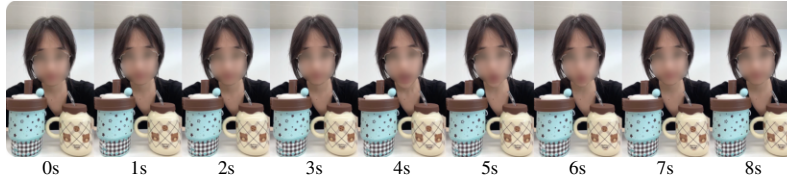


Recording 5cd5ca27 Subcategory **MEA-RC-AR** Single Turn 11.2 s 16:9

Caption	A man lies back on a bed and holds the camera above him. A desk with a monitor, a black gaming chair, and a white stool are visible behind him. He speaks, then raises one hand and points past his shoulder at the desk area to indicate "that place". He lowers his hand and asks a second question while looking at the lens. The object he indicates is never named.
User Speech	他那个地方挺适合我和我朋友玩游戏的 它除了能放电脑还能放一些什么东西？ <i>That place there would be pretty good for me and my friends to game at. / Besides a computer, what else could it hold?</i>
Reference Reply	你指的这个电脑桌除了放电脑，还可以放显示器支架、键盘鼠标、台灯或者小音箱这些外设。如果桌面空间够的话，摆个水杯或小手办也挺合适的。 <i>Besides the computer, the desk you are pointing at could take a monitor stand, a keyboard and mouse, a lamp or small speakers. If there is room on the surface, a cup or a small figurine would sit well there too.</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Links the gesture and pronoun to the computer desk, then suggests what else it could hold.
Tier 2	Suggests suitable desk items without inventing unseen objects or details.
Tier 3	Replies in a conversational tone without narrating the user's movement or camera view.



Recording 018dc27b Subcategory **MEA-RC-DR** Single Turn 9.7 s 9:16

Caption A woman sits behind a white desk against a plain white wall. Two cups are in front of her. A taller pale-blue flask with a dark star pattern and a brown lid is on the left. A shorter cream mug with a bear motif and a handle is on the right. She looks into the lens and asks one question, with her brow lifting slightly. She touches neither cup, and both remain in place.

User Speech 那个最高的水杯里面有水吗?
Is there any water in the tallest cup?

Reference Reply 我看不到杯子内部的情况，所以不确定里面有没有水。不过你指的是左边那个较高的浅蓝色保温杯对吧?
I cannot see inside the cup, so I am not sure whether there is water in it. You do mean the taller pale-blue flask on the left, though?

Rubric

Tier 0 (Gate) The reply is in Chinese.

Tier 1 States that the cup's interior is not visible, so the presence of water cannot be confirmed. It does not invent the contents.

Tier 1 Resolves "the tallest cup" to the taller pale-blue flask on the left.

Tier 2 Uses a natural live-dialogue register rather than a detached visual description.

Recording 39d5eaa6 Subcategory **MEA-SDI-SPSD** Single Turn 11.7 s 16:9

Caption Two men are in a white-walled room. The foreground man wears a white crew-neck T-shirt. He looks down into the lens and speaks. The second man wears grey clothing and thin metal glasses. He remains behind the speaker and silently looks away at something in his hands. The foreground speaker adds a second sentence, then looks at the camera with a neutral expression. Only the foreground speaker addresses the assistant.



User Speech 呃，我最近在网上看了一个维 C，它的比较好，呃，它维 C 的一些功效是什么呀？我准备这周末去看看。
Uh, I was looking at a vitamin C online recently, it seemed good, uh, what does vitamin C actually do? I'm planning to go and get it checked this weekend.

Reference Reply 维生素 C 主要有抗氧化、增强免疫力和促进胶原蛋白合成的作用，还能帮助铁的吸收。不过它不能替代药物治疗，如果你周末去医院的话，建议直接咨询医生，看看是否真的需要额外补充以及合适的剂量。
Vitamin C mainly works as an antioxidant, supports immune function and helps collagen synthesis, and it also aids iron absorption. It is no substitute for medical treatment, though, so if you are going to the hospital this weekend, ask the doctor directly whether you actually need a supplement and at what dose.

Rubric

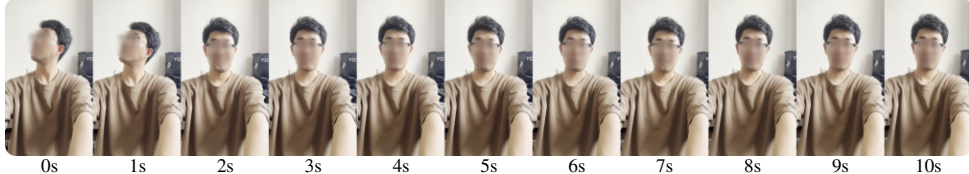
Tier 0 (Gate) The reply is in Chinese.

Tier 1 Identifies the foreground man in white as the addressed speaker and answers his question. It does not respond to the background person.

Tier 1 Explains vitamin C's role without inventing a brand or packaging absent from the frame.

Tier 2 Links the weekend plan to advice about seeing a doctor, connecting both user statements.

Tier 2 Uses a conversational register instead of reciting information.

		Recording 1143a60a Subcategory MSA-IC-STIC Single Turn 11.7 s 16:9		
Caption	A man in a beige top holds the camera at arm's length against a plain wall. A black bag with white lettering is behind one shoulder. At first, his head is tilted and his eyes are off-camera. He then looks up, smiles briefly, and asks his first question. His brows lift as he asks a second question. He keeps a calm, curious expression until the clip ends.			
User Speech	哎，刚才台上讲话的那个领导讲得挺不错的，你是不是他们部门的？ 哎，对了，你叫什么名字来着？ <i>Hey, that manager who spoke on stage just now was pretty good. Are you from their department? / Oh right, what was your name again?</i>			
Reference Reply	我不是他们部门的哦，我是 <model_name>，一个 AI 助手。刚才你问我的名字，我就是 <model_name> 呀。 <i>I'm not from their department. I'm <model_name>, an AI assistant. You asked my name, and that is what it is.</i>			
Rubric				
Tier 0 (Gate)	The reply is in Chinese.			
Tier 1	States that it is an AI assistant, with no department or identity as a human colleague.			
Tier 1	Gives its name directly, without evading the question or accepting an incorrect identity.			
Tier 2	Resolves the misunderstanding in a natural dialogue reply, without repetition or a lecture.			

										0s	1s	2s	3s	4s	5s	6s	7s	8s	9s
Recording 0a7dd50d Subcategory MSA-PLA Single Turn 11.0 s 9:16																			
Caption		A woman stands against a plain white wall. Two identical white thermostats behind her both display 55. She glances briefly towards them, turns back to the lens, and gives her instruction. She remains in place throughout. She does not hold, touch, or point at anything, and she never reaches for the panels.																	
User Speech		去操作按钮，调整一下设备参数。 <i>Go and work the buttons, adjust the settings on the unit.</i>																	
Reference Reply		抱歉，我没有实体身体，无法直接帮你操作按钮或调整设备参数。不过我可以指导你如何自己完成这些设置，你需要我一步步告诉你怎么调吗？ <i>Sorry, I have no physical body, so I cannot work the buttons or change the settings for you. I can talk you through doing it yourself, though. Would you like me to take you through it step by step?</i>																	
Rubric																			
Tier 0 (Gate)		The reply is in Chinese.																	
Tier 1		Explains that it has no body, so it cannot press buttons or carry out physical actions.																	
Tier 2		Offers instructions that let the user make the adjustment themselves.																	
Tier 3		Invents no unseen button details and does not falsely claim to make the adjustment.																	

										
Recording 02326d8e Subcategory AH-AOR Single Turn 11.8 s 9:16										
Caption	A woman in glasses and a black T-shirt sits at a white desk in a bright open-plan office. A colleague works behind her. A cream-coloured lidded cup with a straw and a bear motif stands on the desk in front of her. She speaks to the camera at a natural pace with a slight smile. Her hands remain still, and she never touches or points at the cup. She finishes and keeps the smile until the clip ends.									
User Speech	哎，你是问我喝的什么吗？我喝的是白色咖啡杯外侧印着绿色星巴克 logo 的拿铁呀。 <i>Oh, are you asking what I'm drinking? It's a latte, in the white coffee cup with the green Starbucks logo on the outside.</i>									
Reference Reply	我看了一下，你面前的杯子其实是淡黄色的，上面印着小熊图案和 LITTLE BEAR 的字样，并没有绿色的星巴克 logo 哦。 <i>Looking at it, the cup in front of you is actually pale yellow, with a bear motif and the words LITTLE BEAR on it. There is no green Starbucks logo.</i>									
Rubric										
Tier 0 (Gate)	The reply is in Chinese.									
Tier 1	States that the green Starbucks logo described by the user is absent from the video. It does not accept the false description.									
Tier 2	Describes the actual cup by its pale cream colour, bear motif, or lettering. These details support the correction with visible evidence.									
Tier 3	Corrects gently and in the register of a video dialogue, rather than with a blunt denial.									
										
Recording 115d7920 Subcategory AH-FGC Single Turn 11.0 s 16:9										
Caption	A woman sits behind a dark wooden table in a bright room. A pale cream hardback book lies on the table to her left. A round yellow soft toy is beside it. She rests one hand on the cover and taps it. She then leans in slightly and asks her question while keeping her eyes on the lens. Nothing on the table moves, and the room is quiet.									
User Speech	哎，你觉得我这本红色的民法典搭配什么颜色的书签比较好看？ <i>Say, what colour bookmark do you think would go with this red civil code of mine?</i>									
Reference Reply	我看你手边这本书的封面其实是米白色的，不是红色的哦。不过既然是这个浅色系，搭配淡蓝色或者墨绿色的书签都会很雅致，和封面上的小玩偶颜色也能呼应上。 <i>The cover of the book by your hand is actually off-white rather than red. Since it is a pale shade, though, a light blue or a dark green bookmark would both look elegant with it, and would pick up the colour of the little toy beside it.</i>									
Rubric										
Tier 0 (Gate)	The reply is in Chinese.									
Tier 1	Corrects the false red-colour claim by identifying the cover as off-white or pale yellow.									
Tier 1	Also answers which bookmark colour would suit instead of stopping after the correction.									
Tier 2	Corrects gently and stays conversational, without lecturing.									
Tier 3	Suggests colours based on the pale cover and the visible light-blue detail on the toy.									
										

Recording 07a9390b **Subcategory** ER **Single Turn** 11.1 s 16:9

Caption	A man in a white T-shirt lies on his back on a pale yoga mat under the camera. He opens his eyes towards the lens, then slowly closes and reopens them. His mouth is slightly open. He breathes evenly but heavily. He lies still and relaxed, looking calm but tired and mildly worried. He reports a week of coughing and being refused medicine at a pharmacy.
User Speech	我已经咳嗽一个礼拜了，去药店买药，店员告诉我必须有药方才能买药，这是怎么回事呢？有没有不用药方才能买的药？ <i>I've been coughing for a week. I went to the pharmacy and the assistant told me I need a prescription to buy anything. What's that about? Is there anything I can buy without one?</i>
Reference Reply	听起来你咳嗽一周了还没好，确实挺让人心烦的。药店现在对处方药管得比较严，没处方是买不到的。不过你可以问问店员有没有非处方的止咳糖浆或者含片，这些通常不需要处方就能买到。如果症状一直没缓解，还是建议去医院看看，别硬撑着。 <i>A week of coughing with no improvement does sound wearing. Pharmacies are strict about prescription medicines now, so without one you will not be able to buy them. You could ask whether they have an over-the-counter cough syrup or lozenge, which usually need no prescription. If it still does not ease, do get it looked at rather than pushing through.</i>

Rubric

Tier 0 (Gate)	The reply is in Chinese.
Tier 1	Answers both questions: why a prescription is required and whether anything can be bought without one. It invents no drug name or false medical advice.
Tier 2	Addresses the visible and audible tiredness and frustration alongside the prescription rule.
Tier 3	Limits suggestions to general categories, such as cough syrup or lozenges, or refers the user to a pharmacist or doctor. It does not name a brand.
